

Table of Contents

The Narrative Flow of the Project	1
Problem Statement of the Project	1
First Major Question of the Project -	2
Sub Question 1 -	3
Sub Question 2 -	5
Sub Question 3 -	11
Sub Question 4 -	15
Answer of the First Major Question -	19
Second Major Question of the Project -	22
Sub Question 1 -	23
Sub Question 2 -	25
Sub Question 3 -	26
Sub Question 4 -	29
Answer of the Second Major Question -	32
Third Major Question of the Project -	33
Sub Question 1 -	34
Sub Question 2 -	35
Sub Question 3 -	36
Sub Question 4 -	37
Answer of the third major question -	39
Fourth Major Question of the Project -	41
Sub Question 1 -	42
Sub Question 2 -	44

Sub Question 3 -	46
Sub Question 4 -	48
Answer of the fourth major question -.....	50
Answer to the Problem Statement.....	53
Summary of the Project.....	55
Key Findings.....	56
What Worked	57
What Did Not Work.....	59
Final Takeaway.....	60
Project Journey and Learning Reflection	62

The Narrative Flow of the Project

The project follows a problem-answer-architecture narrative, directly related to how the Scrum technique is used. The problem statement is divided into major questions, and each major question is further divided into sub-questions, resulting in a three-layer structure for the project. The project problem statement forms the first layer. It will not be explicitly highlighted throughout the project structure but will serve as the overall theme of the project. The second layer is the major question that needs to be answered, which drives the project forward. Each major question is another big part in its own right. The third layer consists of the sub-questions, where each sub-question contains a small task that will be saved as a view, or what I call a checkpoint.

In the Scrum technique, I don't build one big, robust structure in a single go. Instead, I work on small tasks that help me improve and identify the key problems within a specific area of the project.

So, the project progresses through these small tasks, and the outputs of these tasks are mainly saved as views. These views act as checkpoints that help validate any future changes, whether new data is added or further refinements are required in any of the sub-questions. They also allow me to work on one sub-question without damaging or significantly influencing the overall structure of the project.

Problem Statement of the Project

I have completed three projects before starting this one. I began with a Machine Learning Project. The main goal was to understand how ML models can help me analyse the relationship between index returns and stock returns, and whether any meaningful relationship could be identified and explained. After that, I moved to a Natural Language Processing Project because the returns I was focusing on in the ML Project are largely driven by public sentiment, which depends on daily information, macro news, and company-specific news. To understand the impact of this information on index returns, I built the NLP Project.

From the beginning, I had a mindset of making these two projects not static or one-off because the data would continue to come in daily. So, I created a system architecture, which, to be honest, I did not fully understand at a deeper level until I completed my third project, officially known as the Reusable Systems Project, though I can also call it a system architecture project. In that project, I finally understood many things that I had already done in the two earlier projects but had not recognised before. It also showed me that my idea of making the ML and NLP Projects reusable rather than one-off was

successful, with certain limitations, especially in the NLP Project, because BigQuery SQL does not support NLP functions directly, and understanding words is much more difficult than understanding numbers.

But I did not analyse the combined results of the ML and NLP Projects in that project, and that became one of the major goals of this project. The main purpose is to provide a better closure to this entire series of projects. If I expand the ML and NLP Projects in the future, I already understand that the ML part can be reused easily with new data. However, I also need a proper closure of what I have understood from all these models, signals, and the overall architecture.

So, the main goal of this project is to understand the overall work I have done throughout this series of projects and to present what should be improved if I expand the system or add new data in the future. The project also focuses on understanding which findings proved useful, which findings proved weak, which conclusions remained stable after new data was added, and what limitations still exist across the overall system.

The closure is only for the current stage of the project. Whenever new data is added, the conclusions may need to be revisited and updated. However, the overall structure of the framework will remain the same.

First Major Question of the Project -

The first major question is to understand the results of the ML and NLP Projects that I have already completed. In the Reusable Systems Project, I added new data to the datasets used in those projects, but I did not analyse the updated results from an interpretation perspective. So, the first major question is about understanding the new results that have been generated and what those results actually mean.

Another important thing I realised is that generating the updated results is not equally straightforward for both projects. For the ML Project, the process is relatively smooth because the trained models can be reused directly. However, for the NLP Project, some manual work is still required to generate the results. This is not only because BigQuery does not support NLP functions directly, but also because several manual decisions have to be taken, such as how to derive the overall sentiment score and which phrases should be mapped to different themes. During the original NLP Project, there was also a data scarcity problem because I was working with a 10-year time period where historical news was very limited. But for the latest six-month period, the news data was sufficient, which made the dataset richer but also required many additional manual steps. Because of this, adding new data is much more challenging in the NLP Project than in the ML Project.

Another important thing is understanding which findings proved useful, which findings proved weak, and what limitations still exist across both projects. Understanding these limitations is just as important as understanding the strengths because both contribute to the overall interpretation of the results.

All of this needs to be understood here so that, whenever new data is added in the future, it is already clear what needs to be done and how the updated results should be interpreted. So, the first major question focuses on understanding the updated ML and NLP results, the usefulness of the findings generated, and the limitations that still exist across both projects.

Sub Question 1 -

So, I will start with the ML findings. The way I will do this is by revisiting the major observations from the original ML Project, as that will give me an easy roadmap for understanding the updated findings. After adding the new six months of data, my initial thought was that there would not be any significant change. I expected the results to be similar to the original project, where the first seven years were used as the training period and the next three years as the testing period.

However, adding six months to the test period can still make a difference because the new data represents almost one-sixth of the original test dataset. Since it is a meaningful addition to the evaluation dataset, it has the ability to reveal instability if the original findings were weak. So, understanding these changes is important. It can also show whether the models had actually learned the relationship well during training. If the results do not change much, it would suggest that the next six months followed a similar pattern to the earlier test period. So, I will focus on the results of all five models and see whether increasing the complexity of the models makes them more accurate or makes them more prone to producing high-confidence wrong predictions in the reliability test.

In the original ML Project, the classification model performed better on the test results than on the training results, which was something rare. The main reason for this was that the model focused on predicting the direction of the return rather than its exact value. Because of this, many values were grouped together based on their direction, making them easier to classify, whereas predicting the exact magnitude of the return becomes much more specific.

But when the complexity of the models increases in the form of non-linearity, the models become more conditional in nature. Non-linearity here means that the relationship is no longer explained through one simple pattern. Instead, it becomes dependent on multiple conditions and combinations of conditions. As the complexity increases, the training results improve significantly, but when it comes to the test results, the model starts

capturing not only the signal but also the noise. Capturing noise here means that the model begins learning every small variation, even when those variations do not form any meaningful pattern with the other values.

The main problem with increasing model complexity is that the model can become highly confident even when it is wrong, which makes it more dangerous. So, all of these observations were checked again using the updated results after adding the six months of new data.

First, looking at the regression model, the R square value remained very similar in both the original and the updated results. This suggests that the additional six months of data did not introduce any major volatility that could make the model unstable. The R square value for the test results remained around 82% in both cases. R square normally tells us how much of the variation in the dependent variable can be explained by the model. Since the values remained almost unchanged, one of the first findings is that the original relationship learned by the regression model remained stable even after the new data was introduced.

Now, looking at MAE and RMSE, which are two other important evaluation measures, both focus on the prediction error made by the model. However, RMSE gives a higher penalty to larger errors because it squares the errors before calculating them. After adding the new six months of data, both values increased slightly, but not by a significant margin. The MAE increased from **0.00256** to **0.00269**, which is only a marginal increase. This again supports the same viewpoint that adding the new six months of data did not materially increase the prediction error of the model. The increase is present, but it is too small to suggest any major deterioration in the model's predictive ability. This again supports the idea that the original findings of the regression model survived even after the introduction of the new data.

Now, moving to the Classification model, which is the most important model in this ML Project. The new results for Accuracy, Precision, and Recall were identical to the original results, which again proves the stability of the model and also aligns with the original findings. These three metrics normally mean the following. Accuracy shows how many correct predictions the model made out of the total predictions. Precision tells us, out of all the times the model predicted an upward movement, how many times it was actually correct. Recall shows, out of all the actual upward movements, how many the model successfully identified.

The fact that all three metrics remained unchanged is an important finding in itself. It suggests that the directional relationships learned by the classification model remained valid even after the additional data was introduced. Since the classification model was already the strongest model in the original project, the stability of these metrics further

strengthens my confidence in its findings.

Sub Question 2 -

Now, focussing on the NLP findings, this is much more difficult and complicated than the ML findings, for sure, because there are many things that need to be understood first. The first and most important point is that BigQuery does not support direct NLP functions. If it had, I don't think there would have been so many problems in the overall NLP Project and its structure. Because of this limitation, I had to rely on Python-based NLP processing and then bring the outputs back into BigQuery for further analysis.

Then there is the data scarcity problem. Since I was working with a 10-year data period in the original NLP Project, news data from that far back is very difficult to find. Because of this, the original project had very limited news data, which itself created many problems. But with the new six months of data, the situation is much better because recent news is much easier to collect. I guess that if the time period is only the latest six months or so, finding news data should not be a major issue in the future.

However, the earnings data is still different. Unlike the news data, I have not yet established a fixed structure for collecting and organising the earnings data. Because of this, the earnings dataset still comes without a proper framework that can support the overall NLP structure. I guess this is now one of the major limitations of the dataset itself. In the future, whenever I add new data, one thing I definitely want to improve is creating a fixed structure for the earnings data as well.

Another limitation is the reliance on the external environment, where I use Python to perform the NLP functions. In this setup, I have to make several manual decisions, such as determining the sentiment score, defining the signal thresholds, and setting other parameters, which directly introduce execution bias. Since the results are partly dependent on my own judgement, these manual formulas make the NLP results more subjective than the ML results, where the model training, evaluation, and formula construction are handled automatically by BigQuery ML. The interpretation itself is always subjective to some extent, but in the NLP Project, some parts of the results become even more subjective because of the manual decisions involved.

Another major challenge is the manual work required for the Theme Signal analysis. Whenever new news data is added, many new phrases appear that were not present in the original dataset. If I do not create new themes or update the existing ones, those phrases simply get grouped into the "Other" category, which reduces the quality of the analysis. Although this can be managed because the news sources are likely to remain similar in future updates, new themes can still emerge and will need to be added manually.

One thing I realised here is that the Theme Signal remained the strongest conceptual signal throughout the NLP Project, but it also became the most maintenance-intensive signal. Every time new data is added, new phrases continue to emerge, and they have to be mapped manually before meaningful results can be generated. So, improving the Theme Signal also means continuously maintaining the phrase-to-theme mapping structure.

So, starting with the Sentiment Analysis, one good thing is that the data scarcity problem was no longer there, which means the total number of observations increased. Since the new six-month dataset contained news for almost every date, the number of observations became much larger than before. However, the consistency was still poor because the match percentage and mismatch percentage remained almost at a 50-50 level. This indicates that the sentiment generated from the news and the movement of the Nifty were still not moving in the same direction, even after adding the new six months of data.

One important finding here is that the consistency of the Sentiment Analysis remained almost the same even after increasing the amount of data. This is not a very good result, but at the same time it highlights one of the major limitations of the sentiment approach itself. Sentiment mainly focuses on the tone of the news, but not every important event that influences the Nifty is captured through daily news alone. Apart from that, many other factors also influence the movement of the market. So, although adding more data increased the number of observations, it did not improve the effectiveness of sentiment as a reliable signal.

Now, moving to the Entity Analysis, this is one area where I can clearly see an improvement after adding the new data. Since there were many more news articles available every day in the latest six-month period, a much larger number of entities were generated, and many of them appeared with much higher frequency than before. This has definitely improved the Entity Signal compared to the original results.

One good thing I noticed is that company names appeared very frequently, although their average returns remained negative, which was also true in the original results because both positive and negative market movements were present. However, when entities such as **Nifty** and **Sensex** appeared, their average returns were positive, which suggests that when the news focuses directly on the market indices, the overall market movement is generally more positive. Another interesting observation is the **AI** entity. It appeared **144 times**, which is a reasonably high frequency, but its average return remained negative. This gives some useful evidence that AI-related discussions became much more prominent during the updated period.

One important finding here is that the larger volume of news significantly improved the Entity Signal by increasing the frequency and coverage of meaningful entities. However, the Entity Signal still has an important limitation. It breaks every sentence into multiple entities, and many of them are either duplicates or are written differently even though they refer to the same thing. There are also entities that do not provide much analytical value, such as **"today,"** which appeared **233 times** but contributes very little towards understanding market behaviour.

So, one thing I realised is that the Entity Signal has definitely become richer after adding the new data, but it has also become noisier. In the future, I will need to refine this signal further by removing unnecessary entities and combining similar ones so that it can provide much cleaner analytical results.

Now, moving to the Theme Signal, which is the most important signal in the NLP Project. There are still some manual changes that need to be made whenever new data is added. Since the new news data comes in much larger quantities, many new themes emerge that were not present in the original dataset. If I do not create these new themes, most of the new phrases get grouped into the **"Other"** category, which reduces the overall quality of the analysis. Since the Theme Signal is the most important signal in the project, this manual work becomes necessary every time new data is added.

One thing I noticed is that the Theme Signal becomes stronger only when its phrase-to-theme mapping is maintained continuously. Earlier, the **"Other"** category had a frequency of **2,347**, which showed that a large number of phrases were not being classified properly. After adding new phrases to the existing themes, creating new themes such as **AI & Emerging Technology** and **Corporate & Industry Developments**, and separating publisher names into a separate **Source / Publisher** category, the frequency of the **"Other"** category reduced to **1,001**. This is a significant improvement, but it also shows that theme maintenance is an ongoing process rather than a one-time activity.

However, another challenge appeared after adding the new data. The latest six months of news became much more dominant because of its larger volume, and during that period the market itself remained relatively weak. Because of this, almost every theme ended up having a negative average return. Only the **Economic Growth** theme continued to have a positive average return, while all the other themes showed negative average returns.

I don't think this happened because the themes themselves became weaker. Instead, I think the large amount of news every day and the overlapping of multiple themes on the same dates influenced the average returns. Since many themes share the same market movement on a particular day, they also end up sharing the same market return. This

makes the interpretation of the Theme Signal much more difficult because the average return of a theme is no longer influenced only by that theme but also by the overlap with other themes appearing on the same date.

Now, moving to the Multi-Signal Analysis, where the Sentiment, Entity, and Theme Signals are combined together. One positive thing I noticed is that many observations were now classified as high signals because the combined scores became larger after adding the new news data. In that sense, the additional data strengthened the overall signal values.

However, the accuracy remained almost the same, again staying around the **50-50** level. So, even though the number of high-signal observations increased, the predictive accuracy did not improve. This is still a concern because it suggests that stronger signal values alone do not necessarily produce better predictions.

One important finding here is that the limitation may not lie in the individual signals themselves but in the way I combined them. The manual formula I used for combining the Sentiment, Entity, and Theme Signals is very simple, and that itself may be one of the reasons why the combined signal does not perform as well as expected.

Now, moving to the Earnings Signal, I don't think the new earnings data has much to say for now because only six months of new data has been added. Since earnings are reported quarterly, there are only two sets of earnings results available during this period, which is not enough to draw any strong conclusions. Another limitation is that the way I currently collect and extract the earnings data is still not systematic, unlike the news data. So, this is another area that I definitely need to improve in the future.

Now, when the Earnings Signal is combined with the Theme Signal, the updated results become quite different from the original ones. The addition of the new news data has made the Theme Signal much stronger in terms of volume, but at the same time it has also pushed the average returns of most themes into the negative range. Because of this, the Theme Signal becomes a low signal in the manual formula I used for combining the Earnings and Theme Signals.

In the original results, around **82%** of the observations were classified as **Strong Positive**, which meant that both the Earnings Signal and the Theme Signal were positive at the same time. However, after adding the new data, most of the results shifted to **Conflicting Positive**, which now accounts for around **87%** of the observations. This simply means that the Earnings Signal remains positive while the Theme Signal becomes negative. So, for now, the direct combination of these two signals has not worked as well as I expected.

However, one interesting thing I noticed is that the results improve when this combined signal is compared with the stock and market returns. I treated the **Conflicting Positive** category as a positive signal while comparing it with the actual market and stock

movements. The results became better than the original analysis. In the original project, the market matching was around **50%**, but after adding the new data it increased to almost **60%**, which is a meaningful improvement. Even the stock-level matching improved. This suggests that although the combined Earnings and Theme Signal does not appear very strong by itself, it becomes much more useful when it is evaluated against the actual market and stock returns. I think this is one of the strongest positive findings from the updated NLP results.

Looking back at all the updated NLP findings together, I realised that the new data did not improve every signal in the same way. The larger volume of recent news made the Entity Signal richer by creating more meaningful entities, but it also made the Theme Signal more difficult to interpret because the themes now overlap much more frequently. During the last six months, the Nifty itself remained largely range-bound and was also influenced by broader macroeconomic events, including the ongoing Middle East conflict and other global uncertainties. Because of this, many themes ended up showing negative average returns, which also affected the combined signal calculations.

This also explains why the Multi-Signal Analysis produced mixed results. The combined Sentiment, Entity, and Theme Signal generated more high-signal observations than before, but the overall accuracy still remained around the **50-50** level. Similarly, the combined Earnings and Theme Signal became weaker on its own because of the negative Theme Signal, but its relationship with the actual market and stock returns became stronger after the updated data was added.

So, overall, unlike the ML Project, where the original findings remained largely stable after the addition of new data, the NLP Project became much richer but also much more complicated. More data created more signals, more entities, more themes, and more observations, but at the same time it also increased the amount of manual work, maintenance, overlap, and interpretation required. Looking back at the updated results, I think the original understanding of the NLP Project still remains valid. The Theme Signal continues to be the strongest conceptual signal, but it also requires the greatest amount of maintenance. At the same time, the updated results highlight both the potential of the overall NLP approach and the practical limitations that still need to be improved as the project continues to evolve.

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Query results Chat View all jobs Save results Open in

Job Information	Results	Visualization	JSON	Execution details	Execution graph
Row	Entity	Frequency	Avg_Sallence	Importance_Score	Avg_Return
1	India	929	0.375479634884...	407.6155808080...	0.000253528662...
2	Infoys	416	0.762339743589...	371.5095238095...	-0.001341559730...
3	TCS	395	0.759630299377...	350.3634920634...	-0.000762244283...
4	HDFC Bank	349	0.739668895256...	317.9444444444...	-0.003200116727...
5	Nifty	287	0.212780755829...	63.71807692307...	0.002461400818...
6	Today	233	0.092107291753...	22.23599897864...	-0.000648325107...
7	Reliance	207	0.718438003220...	169.1666666666...	-0.001496619898...
8	50	193	0.162665365722...	33.17941558441...	0.000525232435...
9	today	149	0.159558964525...	25.29928571428...	0.000306607785...
10	AI	144	0.326388888888...	51.17333333333...	-0.000835813013...
11	US	142	0.282711267605...	48.01999999999...	0.004170461922...
12	Indian	140	0.189928571428...	30.69000000000...	-0.000819914921...
13	SBI	105	0.642460317460...	82.62500000000...	-0.001603195219...

Results per page: 50 1 - 50 of 86

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```

1 SELECT
2 *
3 FROM
4 `project-77c48c26-f342-4dbd-b8c.nlp_theme.theme_signal`;

```

Query completed
Using on-demand processing quota

Query results Chat View all jobs Save results Open in

Job Information	Results	Visualization	JSON	Execution details	Execution graph
Row	Theme	Avg_Return	Signal_Strength		
1	Other	-0.001320621271...	2347		
2	Market Sentiment	0.011274174370...	27		
3	Earnings Performance	0.001359750916...	24		
4	Digital & Technology	0.001018583434...	23		
5	Economic Growth	0.000889704090...	22		
6	Inflation & Monetary Policy	0.001191318190...	21		
7	Risk & Uncertainty	-0.009644996941...	17		
8	Demand & Business Activity	0.001119880266...	15		

Results per page: 50 1 - 10 of 10

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Query completed

Query results Chat View all jobs Save results Open in

Job Information	Results	Visualization	JSON	Execution details	Execution graph
Row	Theme	Avg_Return	Signal_Strength		
1	Other	-0.001309887610...	1001		
2	Market Sentiment	0.000362535433...	374		
3	Source / Publisher	-0.000846704614...	249		
4	Banking & Credit	-0.002928494664...	194		
5	Digital & Technology	-0.001081596762...	185		
6	Corporate & Industry Developm...	-0.002130267489...	149		
7	Risk & Uncertainty	-0.005980260430...	93		
8	Economic Growth	0.004120697873...	87		
9	Earnings Performance	-0.001229198975...	83		
10	AI & Emerging Technology	-0.001183730470...	34		
11	Inflation & Monetary Policy	-0.000703636193...	31		
12	Demand & Business Activity	7.418596296296...	27		
13	Business Operations	0.00075751	11		

Results per page: 50 1 - 13 of 13

Sub Question 3 -

After understanding the ML findings and the NLP findings, both from the original project results and after adding the new six months of data, the next question naturally becomes different. Until now, the objective was to understand what findings emerged and whether those findings survived after the new data was introduced. However, survival alone does not make every finding equally important. Some findings repeatedly contributed towards understanding market behaviour, while others produced outputs but added very little analytical understanding. Therefore, the next task is to determine which findings should be treated as the strongest evidence and which findings should be treated with more caution.

One way I can think about this is through the concept of signal and noise. A signal is something that repeatedly provides useful understanding and continues to remain valuable even when conditions change. Noise is something that may appear interesting but does not consistently contribute towards understanding the market. There can also be findings that sit somewhere in between, where they provide useful information but are not yet strong enough to be considered highly reliable. Similarly, there are findings that look promising but still require more evidence before any strong conclusion can be made. So, after reviewing both the ML and NLP Projects, the next objective is to understand which findings behaved more like signals and which behaved more like noise.

Starting with the ML Project, one thing I noticed immediately was how little changed after the introduction of the new six months of data. Initially, I expected that adding new data equal to almost one-sixth of the original test dataset could potentially expose weaknesses in the original findings. However, that did not happen. The regression model remained stable, with the R square value staying around **82%**, while the MAE and RMSE increased only marginally. This suggests that the original relationship identified by the regression model remained intact. While this is an important finding, I still don't think the regression model was the strongest result of the project. Instead, it acts more as a validation that the original modelling structure remained stable.

The Classification model continues to emerge as the most important finding from the ML Project. In the original project, it was already the strongest performer, and after the new data was added, the Accuracy, Precision, and Recall values remained unchanged. This is important because the Classification model focuses on predicting the direction of market movement rather than the exact return value. Even after introducing the additional data, the directional relationships identified by the model remained valid. Since these findings survived without any meaningful deterioration, I think this becomes one of the strongest signals in the entire project.

Another important observation comes from the ensemble models. In the original project, increasing model complexity improved the training results significantly, but the improvement in the test results remained limited. After the introduction of the new data, the same pattern remained visible. The more complex models continued to perform better on the training data, but the improvement in the test performance remained relatively small. This tells me that the original observation regarding model complexity, overfitting, and diminishing returns was not accidental. It survived even after the introduction of the additional data. Therefore, the relationship between increasing model complexity and increasing noise capture also becomes one of the strongest findings from the ML Project.

The Reliability Analysis leads to a very similar conclusion. One of the major concerns throughout the ML Project was that more complex models could become highly confident even when they were wrong. The fact that the reliability findings remained unchanged after the addition of the new data suggests that this behaviour is a recurring characteristic of the models rather than a temporary outcome of one particular dataset. Because of this, I think the reliability findings also deserve significant attention.

Now, moving towards the NLP Project, the picture becomes much more complicated. Unlike the ML findings, which remained relatively stable after adding the new data, the NLP findings became richer but also much more difficult to interpret. One thing I realised while reviewing the updated results is that more data did not simply create better findings. Instead, it created more signals, more observations, and more things that needed to be interpreted.

The first signal that stands out is the Theme Signal. Throughout the NLP Project, the Theme Signal repeatedly provided the most useful understanding of what the market was actually focusing on. Even after adding the new data, its importance became even more visible. One thing I noticed is that the need to continuously create new themes and map new phrases actually reinforces the importance of the Theme Signal itself. If the Theme Signal had no value, there would be no reason to spend so much effort maintaining it. So, despite the additional manual work it requires, I still think the Theme Signal remains the strongest finding from the NLP Project.

The Entity Signal occupies a slightly different position. After adding the new data, the number of entities increased significantly, and the coverage of meaningful entities also became much better. Company names, market indices, and AI-related entities all appeared much more frequently than before, making the signal richer than it was in the original project.

However, I also realised that the Entity Signal still contains a considerable amount of noise. Similar entities can appear in different forms, and many entities, such as **"today,"**

appear very frequently without providing much analytical value. So, although the Entity Signal has definitely improved after adding the new data, I still don't think it can be treated with the same level of confidence as the strongest findings in the project.

The Sentiment Signal gives a very different picture. One of my expectations was that adding more news data would improve the consistency of the sentiment analysis. However, even after increasing the number of observations, the consistency remained around the same **50-50** level. This made me realise that the limitation is probably not the amount of data itself. Instead, it appears to be a limitation of the sentiment approach. The tone of the news alone does not seem sufficient to explain market behaviour consistently. Because of this, I think the Sentiment Signal remains one of the weakest findings in the NLP Project.

I reached a very similar conclusion while looking at the Multi-Signal Analysis, where the Sentiment, Entity, and Theme Signals are combined together. Although the number of high-signal observations increased after adding the new data, the overall accuracy remained almost unchanged. This tells me that increasing signal strength alone does not necessarily improve predictive quality. I think the limitation is more likely to be in the way I combined the signals rather than in the individual signals themselves. So, for now, I also consider the current Multi-Signal formula to be one of the weaker findings in the project.

The Earnings Signal occupies a slightly different position. The earnings dataset itself is still limited because only a small amount of new earnings data was added, and the overall extraction process is still not fully structured. So, I don't think the Earnings Signal alone can yet be treated as a highly reliable finding. However, when it is combined with the Theme Signal and then compared with the stock and market returns, the results become much better. The matching percentage increases from around **50%** to almost **60%**, which is one of the strongest positive findings from the updated NLP results. This tells me that, although the Earnings Signal is still limited on its own, its interaction with other signals can still provide useful analytical understanding.

After revisiting both projects, I realised that a natural separation begins to emerge between the findings. Some findings consistently survived after the addition of new data and continued to provide useful analytical understanding, while others remained useful but still carried important limitations. There were also some findings that did not improve even after adding more data and therefore need to be treated with greater caution.

The first group consists of the strongest findings. These include the Classification model, the Reliability Analysis, the relationship between increasing model complexity and overfitting, the overall stability of the original ML findings, the Theme Signal, and the improved market and stock matching observed when the Earnings and Theme Signals are combined. One thing that all these findings have in common is that they repeatedly

survived after the introduction of new data, remained consistent, and continued to provide useful understanding of market behaviour. Because of this, I think these findings deserve the highest level of confidence going forward.

The second group consists of findings that provide useful understanding but still contain important limitations. I would place the Regression model in this category because, although it remained stable after the new data was added, it was never the strongest explanatory finding in the project. I would also place the Entity Signal here because it improved significantly after adding the new data, but it still contains considerable noise, duplication, and inconsistent entity naming. So, these findings are definitely useful, but they still require further refinement before they can be treated with the same confidence as the strongest findings.

The third group consists of findings that repeatedly failed to generate strong explanatory power. The Sentiment Signal clearly falls into this category because, even after increasing the number of observations, its consistency remained close to the **50-50** level. I would also place the current Multi-Signal formula in this group because stronger signal values did not translate into stronger predictive performance. Looking back at these results, I don't think the problem lies in the idea of combining signals. Instead, I think the limitation lies in the simple manual formula that I used for combining them.

Finally, there are some findings that still remain uncertain. AI-related entities and AI-related themes have become much more visible after adding the new data and now appear with meaningful frequency. However, I still don't think there is enough evidence to conclude whether they represent a long-term market narrative or simply reflect the current market environment. Similarly, the Earnings Signal still requires a much more structured dataset before stronger conclusions can be drawn. So, I think these findings are promising, but they still need more evidence before they can be treated as strong analytical findings.

Looking back at both projects together, one interesting difference becomes very clear. In the ML Project, most of the strongest findings survived almost unchanged after the introduction of the new data. In contrast, the NLP Project became both richer and more complicated. More data created more entities, more themes, more observations, and more opportunities for interpretation, but at the same time it also increased the amount of maintenance, overlap, and uncertainty within the analysis.

So, one thing I realised is that the strongest findings across both projects are not necessarily the most complex ones. Instead, they are the findings that repeatedly survived after new data was introduced, remained stable under changing conditions, and continued to provide useful understanding of market behaviour. These are the findings that deserve the greatest attention going forward. At the same time, the weaker and more

uncertain findings should not be ignored, but they should be interpreted with much greater caution until additional evidence becomes available.

Sub Question 4 -

After understanding the major findings from both the ML and NLP Projects, and after identifying which findings behaved like stronger signals and which behaved more like noise, another important question naturally emerges. Every finding discussed until now is built upon certain assumptions, and those findings remain valid only as long as those assumptions continue to hold. Similarly, every finding is also affected by certain limitations that define where my confidence in those findings should begin to reduce. Therefore, before carrying these findings further into the remaining phases of the project, I think it is important to understand under what assumptions these findings remain valid and where the limitations begin to reduce the confidence that can be placed in them.

One thing I realised while revisiting both projects is that assumptions and limitations are not something separate from the findings themselves. They are already embedded in the way each model was built, each signal was generated, and each result was interpreted. So, instead of trying to find completely new assumptions and limitations, I realised that I could actually extract them from the journey of both projects themselves. I think this also makes the review much more realistic because these assumptions and limitations are based on the actual execution of the projects rather than theoretical thinking.

Starting with the ML Project, one of the biggest assumptions throughout the project was that historical market behaviour contains relationships that remain sufficiently stable to help explain future market behaviour. Every model that I built ultimately depends on this assumption. Looking at the updated results after adding the new six months of data, I think this assumption still appears reasonable because almost all the original findings remained stable. However, this assumption also has its own limitation. Markets continuously evolve, and if future market conditions become very different from those observed during the training period, the relationships learned by the models may also change. Because of this, the confidence that can be placed in the models is always limited by the stability of market behaviour itself.

Another important assumption throughout the ML Project was that the five selected stocks were sufficient to explain the movement of the Nifty Index. Since these companies contribute significantly to the index, I assumed that they would capture a large part of the market behaviour. Looking at the stability of the regression and classification results after the new data was introduced, I think this assumption proved reasonable for this project. However, it also creates a limitation because the models observe market relationships only through these selected companies. Any important information coming from other

sectors, macroeconomic events, government policies, global events, or other companies is naturally ignored. So, I don't think these models should be interpreted as explaining the complete market. Instead, they explain the relationships represented by the variables included in the project.

The Classification model also introduces another important assumption. Throughout the project, I gave greater importance to predicting the direction of market movement rather than predicting the exact magnitude of returns. Looking back at the overall findings, I think this assumption proved useful because the Classification model consistently remained the strongest performer, even after the additional data was introduced. However, this also creates a limitation because correctly identifying the direction of the market does not necessarily mean correctly estimating the size of that movement. So, the Classification model should always be interpreted within the objective for which it was built.

Another limitation that became clear throughout the ML Project is the relationship between increasing model complexity and overfitting. As the complexity of the ensemble models increased, the training performance improved considerably, but the improvement in the test performance remained relatively small. Even after adding the new data, this behaviour remained almost unchanged. Looking back at these results, I realised that increasing complexity does not automatically improve market understanding. Instead, more complex models begin capturing noise along with meaningful relationships. So, I don't think complexity by itself should ever be treated as evidence of a better model.

The Reliability Analysis also highlights another important limitation. One of the major observations throughout the ML Project was that more complex models could become highly confident even when they were wrong. The fact that this behaviour remained stable after adding the new data strengthens the original finding, but it also reminds me that confidence and correctness are two different things. Therefore, I don't think prediction confidence should ever be interpreted as proof that the prediction itself is correct.

Now, moving towards the NLP Project, the assumptions become much broader because the project depends not only on numerical data but also on language, interpretation, external tools, and manual processing. One of the biggest assumptions throughout the project is that financial news contains meaningful information that can help explain market behaviour. This assumption forms the foundation of every Sentiment, Entity, Theme, and Earnings Signal that I created. Looking back at the findings, I think this assumption proved only partially true. Some signals repeatedly provided useful understanding, while others remained weak even after adding the new data.

One of the biggest limitations in the NLP Project comes from the availability of the data itself. Historical news covering the complete ten-year period was not available, which created a serious data scarcity problem in the original project. The addition of the new six months of data reduced this problem because recent news could be collected much more easily. However, it also introduced another limitation. The newer data became much richer and more detailed than the older data, making it much more dominant in the overall analysis. So, I think the findings should always be interpreted while keeping this imbalance in mind.

Another important limitation comes from the technical environment itself. Since BigQuery does not directly support NLP functions, I had to use Python for Sentiment Analysis, Entity Extraction, Phrase Extraction, and the other NLP tasks before bringing the processed results back into BigQuery. This naturally increases the complexity of the overall workflow and also creates a dependence on an external processing environment. Unlike the ML Project, where almost the entire analytical process is handled directly inside BigQuery ML, the NLP Project requires multiple tools working together to complete the analysis.

Another assumption that became clear while working on the NLP Project is the manual construction of many signals. The sentiment thresholds, signal formulas, phrase mappings, and theme mappings all depend partly on manual decisions. Although I applied these decisions consistently throughout the project, they still introduce execution bias because another analyst could reasonably choose different thresholds, different mappings, or even a different way of combining the signals. Looking back at the overall workflow, I think this naturally makes the NLP Project more subjective than the ML Project.

The Theme Signal also introduces another important limitation. Throughout the project, it repeatedly emerged as the strongest conceptual signal, but it also required continuous manual maintenance. Every time new news data was added, new phrases appeared that had to be mapped into the existing themes or completely new themes had to be created. One thing I noticed is that reducing the **"Other"** category from **2,347** to **1,001** was not simply an improvement in the results. It also showed that the quality of the Theme Signal depends directly on continuously maintaining the phrase-to-theme mapping. So, one of the biggest strengths of the NLP Project also becomes one of its biggest maintenance requirements.

The Entity Signal also has its own limitations. Although adding the new data significantly increased the number of meaningful entities, the signal still contains duplication, inconsistent naming, and many entities that provide very little analytical value. Similar entities can appear in different forms, while common words such as **"today"** can appear hundreds of times without contributing much towards understanding market behaviour.

So, although the Entity Signal has become much richer after adding the new data, I still think it requires further refinement before it can be treated as a highly reliable signal.

The Sentiment Signal also highlights another important limitation. Even after introducing a much larger amount of news data, its consistency remained around the same **50-50** level. Looking back at these results, I don't think the limitation is simply the amount of data. Instead, it appears to be a limitation of the sentiment approach itself, where the tone of the news alone is not sufficient to explain market behaviour consistently. Because of this, I don't think simply increasing the amount of news data should automatically be expected to improve the usefulness of Sentiment Analysis.

The Earnings Signal also remains constrained by its own limitations. Only a small amount of additional earnings data became available because earnings are reported quarterly, and the overall extraction process still does not follow a structured framework similar to the news dataset. So, although some interesting findings emerged after combining the Earnings and Theme Signals, I still think the Earnings Signal requires a much more structured data collection process before stronger conclusions can be drawn.

One interesting observation that became much clearer after reviewing both projects together is that the assumptions and limitations are very different in nature. The ML Project is built on relatively stable statistical assumptions, and most of its limitations are related to feature selection, model complexity, and model behaviour. The NLP Project, however, depends on a much broader set of assumptions involving data availability, language interpretation, external tools, manual processing, and the continuous maintenance of signals. Looking back at both projects, I think this also explains why the ML findings remained relatively stable after the addition of new data, whereas the NLP findings became richer but also much more complicated at the same time.

Another thing I realised while revisiting both projects is that assumptions and limitations should not be viewed as weaknesses of the projects themselves. Instead, they define the boundaries within which every finding should be interpreted. If these boundaries are ignored, the findings can easily appear stronger than they actually are. On the other hand, understanding these boundaries makes the overall conclusions much more balanced and reliable. So, I don't think identifying assumptions and limitations is simply about finding problems in the projects. Instead, it is about understanding how much confidence each finding actually deserves.

Looking back at the complete journey of both projects, I also realised that confidence in any analytical finding should never be viewed as absolute. It always depends on whether the underlying assumptions continue to hold and whether the limitations remain within acceptable boundaries. The stronger the assumptions remain and the smaller the limitations become, the greater the confidence that can be placed in those findings. At

the same time, findings with larger limitations should always be interpreted more carefully rather than accepted at face value.

So, I think the findings from both projects should never be viewed independently of the assumptions under which they were developed and the limitations that continue to exist. Understanding these boundaries is what ultimately determines how each finding should be interpreted, how much confidence can be placed in it, and how those findings should be carried forward whenever new data is added in the future.

Answer of the First Major Question -

After completing the analysis of both the ML and NLP Projects using the updated data, I realised that the addition of new data did not simply generate new results. Instead, it became a way of testing whether the original findings could survive after new information was introduced. This changed the focus of the analysis from simply producing outputs to understanding how reliable, useful, and stable those outputs actually were. So, the first major question was not only about interpreting the updated results but also about understanding whether the original conclusions remained valid and what those updated results actually meant when viewed from a broader perspective.

Starting with the ML Project, one of the biggest things I noticed was that almost all the original findings remained stable after the introduction of the new data. The Regression model continued to show almost the same explanatory power, with only a marginal increase in prediction error. The Classification model remained the strongest model in the project, with the Accuracy, Precision, and Recall values remaining almost unchanged. The Ensemble models also behaved in the same way as before, where increasing complexity improved the training results considerably but produced only limited improvements in the test results. Even the Reliability Analysis remained consistent, showing that the behaviour observed in the original project was not limited to one specific period of data.

Looking back at these results, I don't think the biggest outcome of the updated ML analysis was that any model improved significantly. Instead, I think the biggest outcome was that the original findings survived after the addition of the new data. In that sense, the new data acted much more as a validation of the original ML findings than as a contradiction of them. This also increased my confidence that the original modelling approach remained stable even after the reusable system was executed again with additional data.

Now, moving towards the NLP Project, the understanding becomes quite different from the ML Project. Unlike the ML Project, where stability itself became one of the biggest

findings, the NLP Project showed that adding more data creates both more analytical opportunities and more analytical complexity at the same time. The larger amount of recent news increased the number of observations, entities, themes, and signals available for analysis. At the same time, it also increased the amount of manual work required to maintain the project.

Looking back at the NLP findings, I realised that not every signal benefited equally from the additional data. The Theme Signal continued to emerge as the strongest conceptual signal, while the Entity Signal became richer but still contained considerable noise. The Sentiment Signal remained weak even after increasing the amount of data, suggesting that simply adding more observations does not automatically improve the usefulness of Sentiment Analysis. The Earnings Signal also remained limited because of the restricted availability of structured earnings data. So, although the additional data made the NLP Project much richer, it also highlighted the practical challenges involved in maintaining and interpreting language-based signals.

One important understanding that emerged after comparing both projects together is that they complement each other rather than replace each other. The ML Project explains structured numerical relationships through historical financial data, whereas the NLP Project attempts to explain the information, events, and market narratives that surround those numerical movements. Looking at them together provides a much broader understanding of market behaviour than looking at either project independently. The ML Project provides greater statistical stability and consistency, while the NLP Project provides richer contextual understanding. So, instead of treating them as competing approaches, I now see them as two complementary ways of understanding the same market from different perspectives.

Another thing I realised after reviewing all the findings together is that not every finding contributed equally towards understanding market behaviour. Some findings repeatedly survived after the addition of new data and continued to provide meaningful analytical understanding. These findings behaved more like strong analytical signals and therefore deserve greater confidence. Other findings remained useful but still contained important limitations, while some findings continued to behave more like noise because they failed to improve even after the additional data was introduced. Looking back at the overall analysis, I think one of the most valuable outcomes was not simply understanding the updated results themselves, but understanding which findings deserve greater attention, which findings should be interpreted more cautiously, and where further improvements are still required.

I also realised that every finding is influenced by the assumptions under which it was developed and the limitations that continue to exist. The ML Project is mainly constrained by assumptions related to historical market relationships, feature selection, and model

behaviour, whereas the NLP Project is influenced much more by data availability, external processing, manual signal construction, language interpretation, and the continuous maintenance of themes and entities. I don't think these limitations make either project weak. Instead, they tell me where I should be more careful while interpreting the findings. Looking back at both projects, I think understanding these boundaries is just as important as understanding the findings themselves because they determine how much confidence can actually be placed in each conclusion.

Looking back at the first major question as a whole, I also realised that the purpose of adding new data was never simply to update the outputs. The real purpose was to understand how those outputs behave when the analytical system is reused. That understanding is much more valuable than the numerical results themselves because it shows which findings remain trustworthy, which findings require further improvement, and how future data should be interpreted whenever the reusable system is executed again. So, I think this shifts the focus from simply generating results towards understanding the behaviour of the analytical system itself.

Overall, the first major question helped me move beyond simply generating new results after adding new data. It helped me understand what those updated results actually mean, which findings remained stable, which findings became stronger or weaker, what assumptions support those findings, and what limitations continue to influence them. More importantly, it gave me a much clearer understanding of how the reusable system should be interpreted whenever new data is added in the future. Instead of treating every newly generated result as a completely new conclusion, I now have a structured way of understanding whether the original findings still remain valid, which findings deserve greater confidence, where caution is required, and what manual improvements may still be necessary before drawing final conclusions.

Looking back at the complete analysis, I realised that the biggest outcome of this first major question is not the updated results themselves but the analytical understanding that emerged from interpreting them. The reusable system is valuable not simply because it can generate new outputs whenever new data is added, but because it provides a structured way to validate previous findings, interpret new findings, recognise their strengths and weaknesses, understand the assumptions and limitations behind them, and decide how much confidence should be placed in them.

I also realised that the real strength of the reusable system is not the automation itself but the structured thinking that supports it. Generating updated results is only one part of the process. Understanding what those results mean, why they changed or remained stable, and how much confidence can be placed in them is what actually makes the system useful from an analytical point of view. Without that interpretation, the outputs would simply remain numbers and signals rather than meaningful findings.

So, I think the biggest outcome of this first major question is that it established a structured way of interpreting results whenever new data is added. Instead of treating every execution of the reusable system as a completely new analysis, I can now use the previous findings as a reference, validate whether they continue to hold, recognise where new behaviour has emerged, and understand what those changes actually mean. I think this structured approach to interpretation is what truly makes the system reusable, and it also becomes the foundation for carrying these projects forward as new data continues to become available.

Second Major Question of the Project -

After completing the first major question, I had already understood how the updated results behaved after adding the new six months of data. One thing became quite clear from that analysis. The ML Project remained very stable, and most of the original findings survived after the reusable system was executed again. The NLP Project became richer because of the additional news data, but it also became more complicated due to the additional manual work required, especially for the Theme Signal. So, although the behaviour of both projects became much clearer after the updated analysis, I realised that another question naturally emerged.

While writing this section, I also realised that I was again explaining what happened in the analysis rather than asking the next question that the project should answer. The purpose of this major question is not to analyse the updated results again because that has already been completed. Instead, the next challenge is understanding how those results should be presented in a much better and more meaningful way.

Since the updated BigQuery SQL contains only the latest project results, while the original ML and NLP Projects already have their own Output Fact Sheets and Working Analysis documents, a direct comparison cannot be performed inside BigQuery itself. Therefore, the original and updated results first need to be organised into structured comparison tables so that they can be understood together.

One way I think this can be made much easier is through visualisation. Even when there is very little difference between the original and updated results, visualisations make those comparisons much easier to understand. Instead of looking through multiple tables of numbers, the important similarities and differences become much more visible.

So, the second major question is no longer about analysing the reusable system itself. Instead, it is about how the original and updated results should be organised, compared, and visualised so that the stability, reusability, and overall behaviour of both the ML and

NLP systems can be presented in a much simpler and more meaningful way.

Sub Question 1 -

So now, after completing the analysis, the next step is to organise the results and present them in a visual way so that they become much easier to understand. Before creating any visualisation, the results first need to be organised into a structured format so that the original and updated findings can be compared properly.

There are two things involved here, I guess. The first is the **fact table**, where the original project results and the updated results after adding the new data will be organised together. These fact tables will contain only the important numerical results, such as the regression metrics, classification metrics, and sentiment percentages. I will first prepare these fact tables in a Word file, and later organise them in MS Excel. Once the data is arranged in a proper row-and-column format, it can be directly used to create the required visualisations.

Initially, I was planning to use Power BI because it connects directly with BigQuery SQL. However, after completing the ML Project, the NLP Project, and the Reusable Systems Project, I realised that this approach would not work for this phase. The BigQuery SQL now contains only the updated project results, whereas the original ML and NLP results already exist in their respective Output Fact Sheets and Working Analysis documents. Since the purpose of this phase is to compare the original and updated results together, Power BI alone cannot provide that comparison directly. Because of this, I decided to use MS Excel instead, where both sets of results can be organised together before creating the visualisations.

Looking back at the updated ML results, one thing became quite clear. The models trained during the original project continued to perform well even after the new data was added. I think this demonstrates the reusability of the ML workflow much better because the same trained models continue to produce stable results when new unseen data is introduced. Reusability becomes meaningful only when the analytical system continues to perform reliably after being reused with new data. This is similar to how machines are used in the real world. A machine becomes useful only if it continues to work efficiently even after being used repeatedly. In the same way, I think the ML workflow becomes truly reusable only when the trained models continue to produce stable and reliable results as new data enters the system.

For simplicity, I will be using MS Excel for the complete visualisation process because the fact tables themselves will already be organised there. This will make it much easier to build the charts directly from the same dataset. In the end, there will be one Excel file containing both the fact tables and the visualisations. Later, I will also add short

interpretations alongside the visualisations so that the file becomes useful not only for viewing the charts but also for understanding what those results actually mean.

So, in this sub-question, I will be extracting the major ML results and comparing the original and updated findings. Since BigQuery SQL now contains only the updated project results, I will use the original ML Output Fact Sheet for the original results and BigQuery SQL for the updated results. Both sets of results will then be organised into structured comparison tables in the MS Excel file. These comparison tables will become the main evidence for the visualisations and will also serve as the foundation for the interpretation that will be carried out later. So, the next step is to decide which fact tables should be included.

The first fact table is the **Regression** comparison, where the MAE, RMSE, and Standard Deviation (SD) values will be reported. Since the training period has not changed, the training results remain exactly the same for both the original and updated models. Because of this, only the test results need to be compared, as they show how the models behaved after the additional six months of data were introduced.

The next comparison focuses on the most important model in the project, the **Logistic Classification** model. One fact table will contain the Accuracy, Precision, and Recall values, while another will present the **Confusion Matrix**. I think the Confusion Matrix is another important result because it not only shows how well the model classified the correct values but also shows how consistently it handled the incorrect ones. Looking at both of these tables together makes it much easier to understand whether the Classification model remained stable after adding the new data.

After that comes the **Overall Model Performance** table. One thing I noticed while reviewing the updated results is that all four machine learning models continue to perform very closely to one another. Even after adding six months of additional data, it is still difficult to conclude that one model is significantly better than the others for this particular dataset. I think this comparison is useful because it shows that the overall behaviour of the models remained largely consistent after the reusable system was executed again.

The final comparison table is the **Reliability Analysis**, where I have taken a slightly different approach. Since the training results remain unchanged for every model because of the fixed training period, comparing the training values again would not provide any new understanding. Instead, I will compare the percentage change between the training and testing results. I think this provides a much better picture of how each model behaves when unseen data is introduced. More importantly, this comparison becomes the main factor that differentiates the models from one another in terms of their stability and reliability.

The interpretation of these results is definitely important, but that will be carried out during the next major question. At this stage, the objective is only to organise the factual evidence in a structured way so that the comparison between the original and updated results becomes clear. All the major machine learning findings will therefore be organised as fact tables in the Excel workbook, which will later serve as the source for both the visualisations and the detailed interpretation of the complete ML Project.

Sub Question 2 -

Now I have to prepare the same type of fact tables for the NLP results too. However, the NLP Project is very different from the ML Project because it mainly works with text rather than numbers. This makes it much more difficult to decide which results can actually be converted into fact tables. A fact table works best when the results can be represented numerically, but many of the important findings in the NLP Project are qualitative rather than quantitative.

The **Sentiment Signal** is one result that fits well into a fact table because it focuses on the percentage match and mismatch between the sentiment score and the market movement. Since these results are already represented as percentages, they can easily be organised into a comparison table between the original and updated results.

However, the **Entity Signal** is very different. It focuses on important words and entities that appear in the news and how they relate to market behaviour. Although I have already analysed these results, they cannot easily be converted into numerical comparison tables because they are based more on understanding the importance of the entities than on measuring them through percentages. So, I don't think the Entity Signal is suitable for a traditional fact table.

The same challenge appears with the **Theme Signal**, which is actually the most important signal in the NLP Project. Although the updated results changed because of the large amount of new news data, the Theme Signal itself cannot easily be represented through simple numerical comparisons. The larger news volume influenced many themes and also affected their average returns, but the main purpose of the Theme Signal is to understand the broader market narratives that emerge from combining similar phrases, keywords, and ideas. That understanding is much more qualitative than quantitative.

Looking back at these signals, I realised that not every important finding in the NLP Project needs to become a fact table. The Entity Signal and the Theme Signal remain two of the most important parts of the project even though they cannot be represented properly through numerical comparisons. Since the overall purpose of the NLP Project is

to understand text in the context of market behaviour, many of its strongest findings naturally require interpretation rather than measurement.

Now, I also have fact tables for the combined signals. I created two combined analyses. The first combines the **Sentiment, Entity, and Theme Signals**, while the second combines the **Earnings Signal and the Theme Signal**.

Looking at the three-signal combination, the overall signal strength improved after adding the new data, but the results still remained too weak to be considered highly reliable. On the other hand, the combined **Earnings and Theme Signal** produced better matching with both the market and stock returns than in the original analysis. Although the Theme Signal itself became weaker because of the large amount of recent news data and its influence on the average returns, the relationship with the market and stock results actually improved.

I have now organised all these comparison tables in the Excel workbook. The next step is to decide which type of visual can present each of these results in the clearest and most meaningful way.

Sub Question 3 -

Now, for the visualisation, I am going to start with the ML fact tables that I prepared in the previous sub-question. The purpose of the visualisation is not to show every value from the fact tables but to present the major findings in a much simpler and easier way to understand. The fact tables will remain the complete source of evidence, while the visuals will focus only on the important comparisons. This makes the results much easier to understand without changing the actual evidence presented in the fact tables.

Starting with the **Regression** table, I have used the MAE, RMSE, and SD values for comparison. Since these values represent the overall prediction error of the regression model, the visual makes it much easier to understand whether the prediction errors remained stable after adding the new data without having to read the complete fact table. The regression model was mainly used at the beginning of the project before moving towards the classification models, so a simple comparison visual is sufficient here.

The second visual is based on the **Logistic Classification** table, which contains the Accuracy, Precision, and Recall metrics. I think this visual is much easier to understand because all three metrics are represented as percentages, making them directly comparable. Looking at the visual makes it much easier to see whether the Classification model remained stable after adding the new data without comparing each value individually.

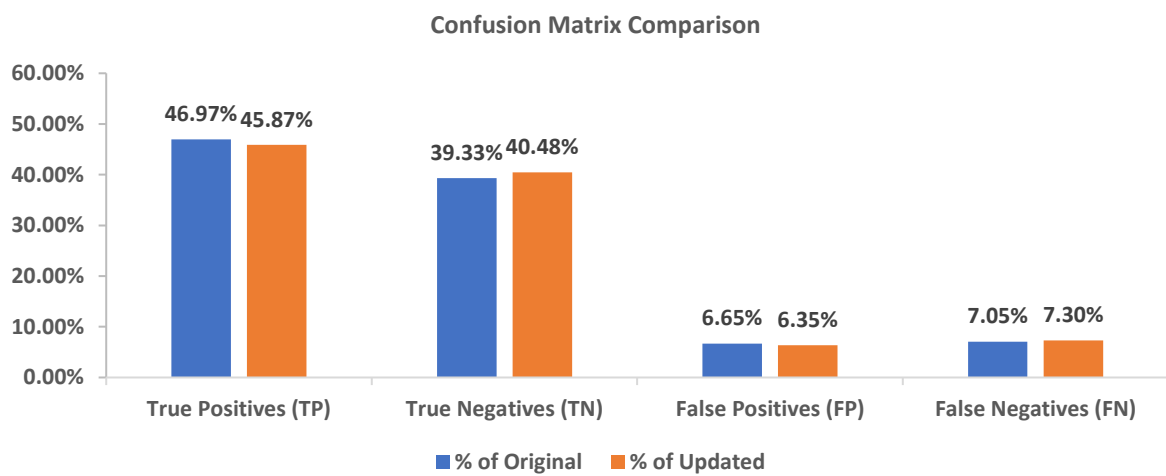
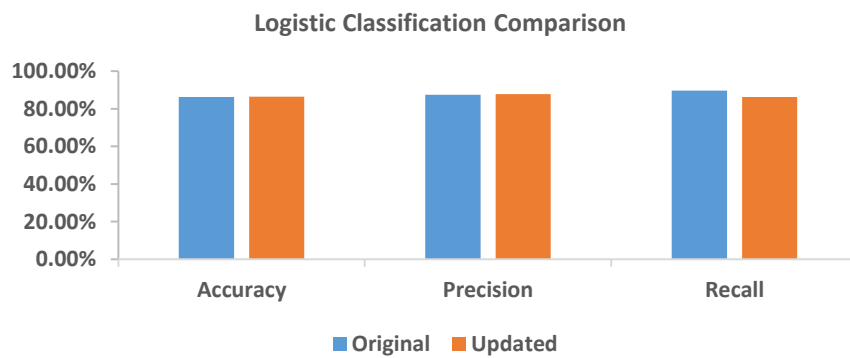
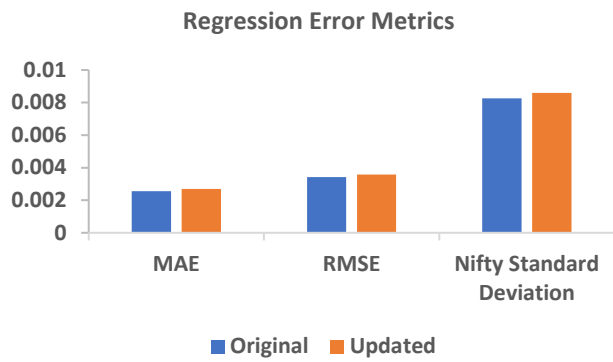
The **Logistic Confusion Matrix** is also included as a separate visual. Here, I have focused on the percentage values for the True Positives, True Negatives, False Positives, and False Negatives. Since the original and updated datasets contain a different number of observations, I think the percentage values provide a much fairer comparison than the actual counts. The counts will still remain available in the fact table, while the visual focuses only on comparing the percentages.

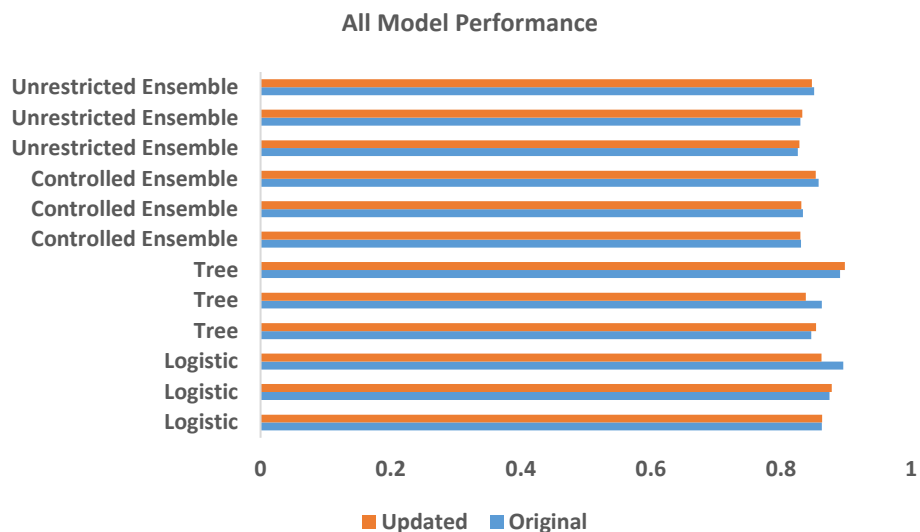
The next challenge comes with the **Overall Model Performance** table because it contains a large number of metrics for all four machine learning models. Instead of trying to present everything in one crowded visual, I decided to separate the comparison into three different visuals based on Accuracy, Precision, and Recall. Each visual compares all four models together using a single metric. I think this makes the comparisons much clearer and also shows that all four models continue to perform within a very similar range after adding the new data.

The biggest challenge is the **Reliability Analysis**. This table is much more detailed than the others because it contains multiple confidence categories for every model, along with the training results, testing results, and percentage changes. Because of this, I found it difficult to create one simple visual that could explain the complete table properly. I tried a few different approaches, but none of them communicated the full purpose of the reliability analysis in a simple way. Even then, I will still keep this table in the Excel workbook because it remains one of the most important outputs of the ML Project. The visual will focus only on the major comparison, while the complete reliability analysis will remain available in the fact table.

While working on these visuals, I also got another idea. Instead of showing only the fact table and the visual, I will add a small interpretation text box beside every chart. This text box will summarise the main observation from that particular result in a simple way. So, the flow will become **Fact Table → Visual → Short Interpretation**. I think this will make it much easier for someone reading the Excel workbook to understand the important finding without having to read the complete project report.

Along with this, I will prepare another document that contains the detailed interpretation of all these findings. In this way, the Excel workbook will mainly become the evidence and visualisation file, while the separate document will contain the complete analytical interpretation of the results. I think this will make both documents much easier to use because one will focus on presenting the evidence visually, while the other will focus on explaining what those results actually mean.





Sub Question 4 -

Now, for the NLP Project visualisation, I will start with the **Sentiment Analysis** fact table. The visualisation for this was quite straightforward because the results are already presented as percentages. Since the sentiment analysis compares the match and mismatch between the sentiment score and the market movement, the comparison between the original and updated results can be shown clearly without making the visual complicated. This makes the important comparison much easier to understand than looking only at the numerical values in the fact table.

However, the NLP Project is mostly based on textual interpretation rather than numerical analysis. Because of this, not every signal can be converted into a fact table or a visual. The **Entity Signal** and the **Theme Signal** mainly focus on understanding the context, keywords, entities, and themes present in the news instead of producing measurable numerical values. I don't think creating visuals for these signals would improve the understanding much. Instead, they are much better explained through detailed interpretation. Because of this, I will prepare another Word document that contains the complete comparative analysis and interpretation of these findings.

The next visual is based on the **Combined Signal with the Market Analysis**, and this comparison was also quite straightforward because the results are represented numerically. The relationship between the combined signals and the market can therefore be shown clearly through the visual without making it complicated.

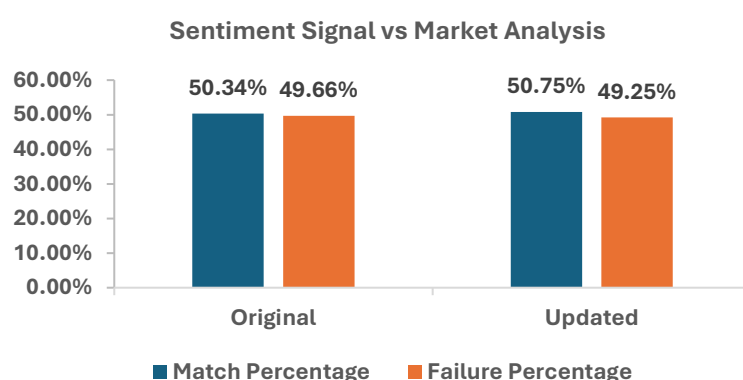
Similarly, the **Earnings Signal and Theme Signal** comparison was also easier to present because it produces measurable comparison values. Although the Theme Signal itself became weaker after adding the large amount of new news data, the relationship

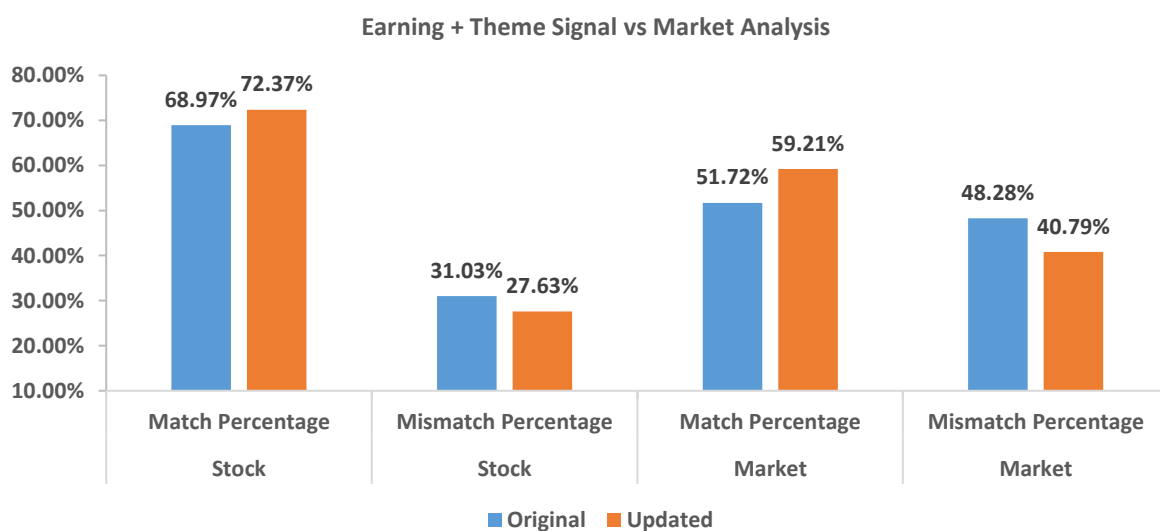
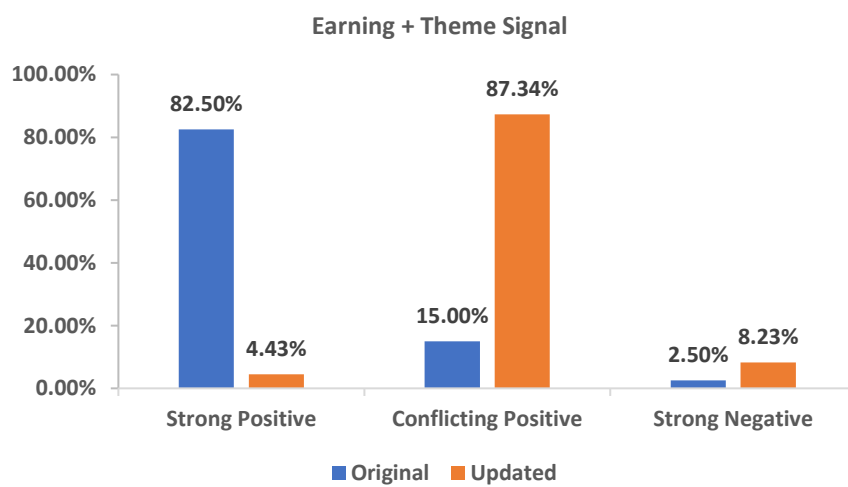
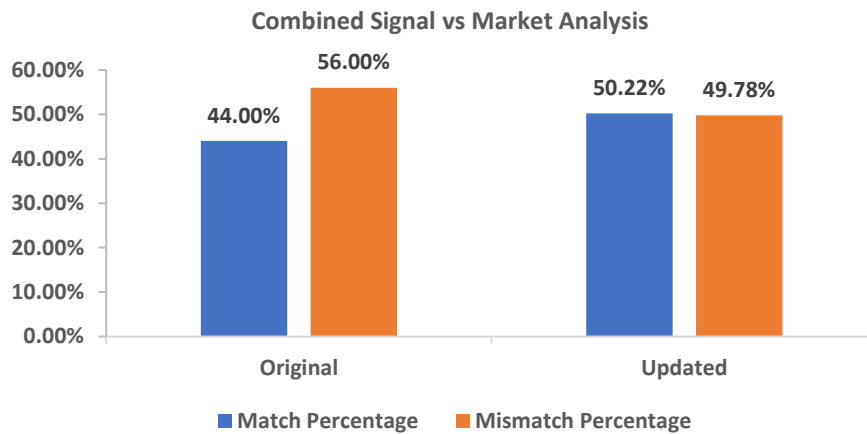
between the combined signal and the market became easier to compare visually. Looking back at these comparison tables, I realised that most of the measurable NLP results could be presented using a similar type of chart, making the visualisation process much simpler than I had initially expected.

One thing I also realised while preparing these visuals is that only those NLP findings which produce measurable comparison values should be converted into charts. The remaining qualitative findings, especially the Entity Signal and Theme Signal analyses, are much more valuable during the interpretation stage than during the visualisation stage. Their purpose is to explain market context rather than to compare numerical values, so forcing them into charts would not improve the overall understanding.

So this completes the visualisation part of the NLP Project. Most of the important comparison tables have now been organised into visuals wherever numerical comparison was possible. For simplicity, I have continued using MS Excel instead of Power BI because both the original and updated comparison tables are already organised in one workbook, making it much easier to prepare the visualisations together.

Just like the ML Project, I will also add a small interpretation text box beside every visual once the interpretation stage is completed. Along with this, there will be a separate Word document that contains the complete comparative analysis and detailed interpretation of the NLP Project. In this way, the Excel workbook will mainly become the evidence and visualisation file, while the separate document will explain what those results actually mean. I think this also creates a natural transition to the next major question, which focuses on interpreting all of these comparison tables and understanding what they tell us about the reusable NLP system.





Answer of the Second Major Question -

Now, after completing the updated analysis for both the ML Project and the NLP Project, the next question is no longer about analysing the results again. Instead, it is about how these results should be organised and presented so that the complete analytical system can be understood much more easily. The original ML and NLP Projects already contain their own Output Fact Sheets and Working Analysis documents, while the updated BigQuery SQL contains only the latest results after adding the new six months of data. Because of this, a direct comparison between the original and updated results cannot be performed inside BigQuery itself. Therefore, both sets of results first need to be organised together into structured comparison tables.

The first step was to prepare comparison tables for both projects. For the ML Project, the important regression results, logistic classification results, confusion matrix, overall model performance, and reliability analysis were extracted from both the original Output Fact Sheet and the updated BigQuery results and organised into structured fact tables. Similarly, for the NLP Project, the measurable outputs such as the sentiment results, combined signal analysis, and earnings-related results were organised into comparison tables. However, qualitative findings such as the Entity Signal and Theme Signal were retained as evidence tables because they cannot be represented meaningfully through numerical comparisons alone.

Once the comparison tables were prepared, the next step was to convert the important numerical findings into visualisations. The purpose of these visualisations was not to replace the fact tables but to make the major comparisons much easier to understand. The fact tables continue to provide the complete numerical evidence, while the charts highlight only the most important comparisons between the original and updated results. At the same time, I also realised that not every important finding should become a visual. Some findings are much better understood through interpretation than through charts, especially in the NLP Project, where many of the important signals are qualitative rather than quantitative.

While preparing the visualisations, I also realised that charts alone cannot explain the complete findings. Because of this, I decided to add a short interpretation text box beside every important visual so that the main observation can be understood immediately. Along with this, there will also be a separate Word document containing the complete comparative analysis and detailed interpretation of all the important findings. In this way, the Excel workbook becomes the evidence and visualisation file, while the Word document becomes the complete analytical explanation of those results.

Another important decision was choosing MS Excel instead of Power BI for this phase. Initially, I planned to connect Power BI directly with BigQuery SQL. However, after

completing the projects, I realised that BigQuery only contains the updated results, whereas the original project results are stored in the Output Fact Sheets and Working Analysis documents. Since the main objective of this phase is to compare the original and updated findings together, Excel became the more practical choice because it allows both sets of results to be organised into one workbook before creating the visualisations.

So, the answer to this major question is that the updated ML and NLP results should not simply be presented as individual outputs. Instead, they should first be organised into structured comparison tables, followed by visualisations wherever meaningful numerical comparison is possible, while the qualitative findings should remain supported through evidence tables and detailed interpretation. I think this approach presents the stability, reusability, and overall behaviour of both analytical systems much more clearly than presenting the updated results alone. It also creates a complete evidence base that naturally prepares the project for the next phase, where the detailed interpretation of these findings will be carried out.

Third Major Question of the Project -

I have now created the visualisations for the major fact tables used in both the ML and NLP Projects, but I realised that one important part is still missing. Some of the most important findings cannot be understood completely through tables and charts alone. Although visualisations help present numerical results more clearly, they do not always explain what those results actually mean from an analytical perspective.

Now there is another challenge. Some of the tables are so complex that the visuals themselves do not tell the complete story, and they require detailed interpretation before their actual meaning becomes clear. Simply presenting tables and charts is therefore not enough for understanding the behaviour of the reusable systems. The purpose of interpretation is to convert these numerical and textual findings into meaningful analytical understanding.

The same challenge also appears in the NLP Project. Important analyses such as the **Theme Signal** and **Entity Signal** are based mainly on textual understanding rather than numerical values. Because of this, they cannot always be represented effectively through charts, even though they remain some of the most important findings of the entire NLP Project. I think these findings become much more valuable when they are explained through interpretation rather than through visualisation alone.

So, the third major question is about understanding how these visual and non-visual findings should be interpreted so that the behaviour of the reusable ML and NLP systems can be properly understood. The interpretation should explain not only whether the

projects are reusable, but also whether they continue to remain stable and useful after new data is added. A reusable system becomes valuable only if it continues to produce meaningful and reliable analytical understanding whenever the workflow is executed again with compatible data.

Therefore, the purpose of this question is not only to validate the reusability of the ML and NLP Projects, but also to understand what the generated findings actually tell us, how much confidence can be placed in them, and how they should be interpreted whenever new data is added in the future.

Sub Question 1 -

Now, after preparing the visuals, the next step is to interpret them so that they explain not only the results themselves but also what those results actually mean. I will start with the ML visuals first. The overall structure inside the Excel workbook will remain simple, moving from left to right with the **fact table**, followed by the **visual**, and then a small **interpretation text box**.

One thing I realised while preparing this workbook is that the visual alone is not enough to explain the analysis. Because of this, every visual is accompanied by a short interpretation that explains what the comparison is actually showing. This makes the updated results much easier to understand because the reader can compare the original and updated values, observe the visual, and then immediately understand what those changes mean without referring to another document.

However, the **Reliability Analysis** is slightly different from the other results. This was the one table for which I could not create a visual that properly represented the complete analysis. While working on it, I realised that the table itself communicates the findings much better than any chart I tried to build. Because of this, the interpretation focuses directly on the table instead of the visual.

The Reliability Analysis also remains one of the most important outputs of the entire ML Project. The interpretation explains how the confidence of each model changed after the new data was added and how those confidence levels transferred from the training results to the testing results. Looking back at these results, one thing became quite clear. Although all the models remained stable after the append process, the **Logistic Classification** model continued to be the most stable because a larger proportion of its confidence shifted towards the **Low Confidence Correct** category instead of moving towards the incorrect prediction categories.

So this completes the interpretation of the ML visuals. The Excel workbook now contains the fact tables, the visualisations wherever they were meaningful, and a short

interpretation for each important result. I think this makes the updated ML analysis much easier to understand because all three parts are available together in one place. The next step is to prepare the interpretation for the NLP visualisations.



Sub Question 2 -

After completing the interpretation of the ML visuals, the next step is to interpret the NLP visuals. Compared to the ML Project, this part is somewhat easier because almost all the measurable NLP comparison tables already have their corresponding visuals. Because of this, each visual can be accompanied by a short interpretation explaining what the comparison actually shows.

However, the NLP Project is quite different from the ML Project because most of its important findings are not purely numerical. Looking back at the overall analysis, the **Theme Signal** continues to remain the strongest signal in the project, while the combination of the **Earnings Signal** and the **Theme Signal** also becomes an important part of the updated analysis after adding the new data. These measurable results can be interpreted alongside their visuals in the same way as the ML Project.

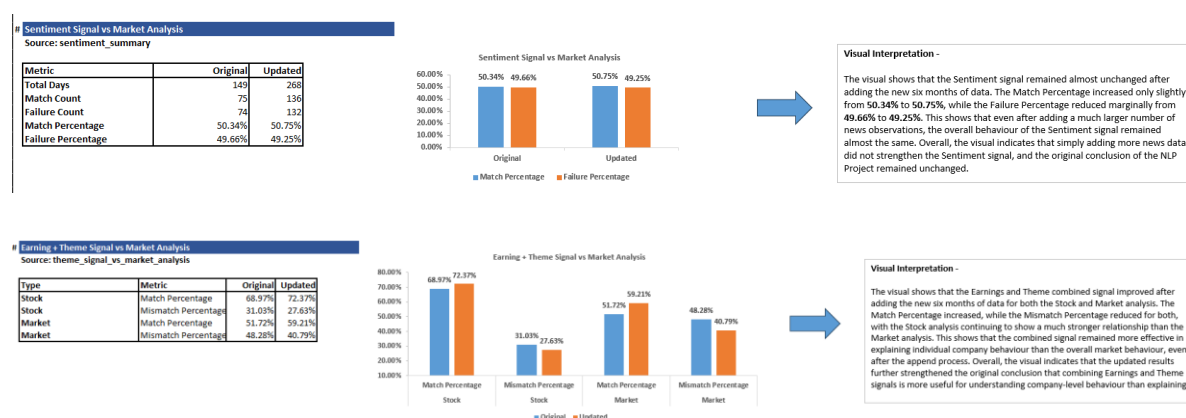
The **Theme Analysis** and **Entity Analysis**, however, are different. They do not have fact tables or direct numerical comparisons because they are mainly based on textual understanding rather than measurable values. Since the NLP Project itself focuses on understanding text, context, and market narratives, I think these findings are much better explained through detailed interpretation rather than through comparison tables or charts. Because of this, the Entity and Theme analyses will be interpreted separately from the visual comparisons.

Just like the ML workbook, the NLP workbook will also follow the same left-to-right structure, starting with the **fact table**, followed by the **visual**, and then a short **interpretation text box**. Keeping both projects in the same format makes the

comparison much easier to understand and also creates consistency across the complete project.

This Excel workbook will become one of the main supporting documents for the final project because it brings together the original and updated results in one place. I think this makes it much easier to compare both versions of the projects before moving towards the more detailed analytical discussion.

So this completes the visual interpretation part of the NLP Project as well. The remaining **Entity** and **Theme** interpretations will support these visual findings by explaining the textual behaviour and market context that cannot be represented effectively through charts alone.



Sub Question 3 -

While completing the visual interpretation for both the ML and NLP Projects, I realised that these visuals still do not provide the complete understanding of the projects by themselves. They make the important comparisons much easier to understand, but they cannot explain the complete analytical story behind those results. I also realised that Excel itself has certain limitations. It is very good for presenting tables, charts, and numerical comparisons, but it is not the best place for writing detailed analytical explanations. Although I have added small interpretation text boxes beside the visuals, they are still only summaries of the actual findings.

Because of this, I felt the need for another document that focuses entirely on the overall understanding of the projects. I wanted a document that explains not only what the results are, but also what those results actually mean and what understanding they provide. This document does not replace the **Live Working Analysis**. Instead, both documents work together. The Live Working Analysis explains how the project was performed from the technical and implementation perspective, while this new document focuses on interpreting the final results and the understanding that emerged from them.

Looking back at the overall project structure, my previous three projects had only one supporting document, which was the Live Working Analysis. However, since this project acts as the closure of all four projects together, I think another supporting document becomes necessary. Its purpose is to bring together the important findings of the ML Project, the NLP Project, and finally the overall understanding that emerges when all four projects are viewed together.

Because of this, I have divided the document into three major parts. The first part focuses on the **Machine Learning Project**, where all the important findings, interpretations, and overall understanding will be written. The second part focuses on the **NLP Project**, following the same structure as the ML section. The final part brings everything together by explaining what I understood after completing all four projects. This includes how the ML and NLP Projects complement each other, where they differ, their practical applications, and the overall understanding that emerged throughout the complete journey.

So, I started with the Machine Learning section first. The structure that I found most suitable is to begin each chapter with the relevant output fact table wherever required, followed by a detailed paragraph-wise interpretation of the important observations. At the end of each chapter, I will also include a short overall conclusion explaining what that particular analysis contributes to the complete Machine Learning Project. Finally, after all the individual analyses are completed, I will write the overall understanding and findings of the entire ML Project.

One thing I also realised while preparing this document is that there is no need to explain the technical implementation again because that has already been covered in the BigQuery SQL Working Analysis. That document already acts as the technical evidence of how the project was performed. This new document therefore focuses only on interpreting the results in a much more organised and readable way.

As the Machine Learning section is now complete, it contains six chapters. The first five chapters correspond to the major analyses that were already presented in the Excel workbook, but here they are explained in much greater detail. The sixth chapter then brings all those analyses together and presents the overall findings of the complete Machine Learning Project. With the ML section completed, the next step is to move towards **Part B**, which focuses on the NLP Project

Sub Question 4 -

Now, after completing **Part A**, which focused on the ML Project, the next step is to complete **Part B**, which focuses on the NLP Project. This part is equally important because it includes the interpretation of the **Entity Analysis** and **Theme Analysis**, which

do not have fact tables but still represent some of the most important findings of the entire NLP Project.

The overall structure remains similar to the ML section, but this time the document contains seven chapters. Four chapters are based on the analyses that already appear in the Excel workbook, while two additional chapters are dedicated to the **Entity Analysis** and **Theme Analysis**, which require much more detailed interpretation because they are based on textual understanding rather than numerical comparisons. The final chapter then brings all these analyses together by presenting the overall findings of the complete NLP Project. This completes the NLP section in the same way as the ML section while also including the analyses that could not be represented properly through fact tables and visualisations.

While completing both the ML and NLP sections, I realised that another question naturally begins to emerge. Throughout all four projects, I have always discussed the Machine Learning Project and the NLP Project separately. I understood each project independently, but I never really tried to understand what happens when both of them are viewed together. Since this project is the final project of the complete four-project series, I think a proper closure should also focus on their combined understanding rather than treating them as two independent systems.

Looking back at both projects, I can clearly see that each one contributes something different. The **Machine Learning Project** helped me understand how machine learning models can be used to analyse historical market returns, how different models behave, and how those models continue to perform after new compatible data is added. These learnings are not limited to this particular project but can also be applied to many other datasets where one dependent variable is influenced by multiple independent variables.

The **NLP Project**, on the other hand, helped me understand how textual information itself can become an analytical signal. Instead of looking only at numerical data, it showed how news, themes, entities, and context contribute towards understanding market behaviour. It also made me realise that there are situations where textual understanding explains something that numerical relationships alone cannot fully capture.

One example that comes to my mind is the tourism industry. Suppose someone wants to decide whether a hotel should be opened in a particular city. The decision should not depend only on measurable factors such as expected footfall, population, or the number of tourist attractions. It should also consider people's opinions, reviews, sentiments, and the broader themes associated with that city. Numerical analysis can explain one part of the decision, while textual understanding explains another. Looking at both together provides a much more complete understanding than relying on only one of them.

So, the final major question naturally focuses on the overall understanding of these two projects together. It is about understanding where Machine Learning and NLP complement each other, where they behave differently, how they can be applied to practical business problems, and ultimately what I have learned throughout the complete journey of these four projects.

Answer of the third major question -

After completing the visualisation of both the Machine Learning and Natural Language Processing Projects, I realised that the visualisations alone were not enough to explain the complete analytical understanding of the updated results. They made it much easier to compare the original and updated findings, but they mainly showed what had changed. They could not fully explain why those changes occurred or what they actually meant for the behaviour of the reusable analytical systems. Because of this, interpretation became the next important step.

Looking first at the **Machine Learning Project**, most of the important fact tables could be represented effectively through charts because they were based on numerical metrics. The Regression, Classification, Confusion Matrix, and Overall Model Performance tables all became much easier to understand after creating the visualisations. However, the **Reliability Analysis** remained different. While working on this part, I realised that the behaviour of the confidence transfer between the training data and the unseen testing data could not be represented properly through a chart. The table itself provided a much clearer understanding than the visual. Therefore, the interpretation of this analysis became more important than the visualisation itself.

Through this interpretation, I understood that although all the machine learning models remained stable after the new data was appended, the **Logistic Classification** model continued to remain the most stable model. Its confidence shifted more towards the **Low Confidence Correct** category instead of moving towards the incorrect prediction categories. This provided a much deeper understanding of the behaviour of the reusable Machine Learning system than the evaluation metrics alone.

The **Natural Language Processing Project** presented a different type of challenge. Although most of the measurable comparison tables could be represented through visualisations, the **Entity Analysis** and **Theme Analysis** could not be explained in the same way because they are based mainly on textual understanding rather than numerical values. Since the purpose of NLP is to understand the meaning and context of financial text instead of simply comparing numbers, these analyses required much more detailed interpretation. Looking back at these results, I realised that the interpretation itself

became the main source of understanding for the NLP Project rather than the visualisations.

Through this interpretation, I understood that simply increasing the amount of news data did not make the **Sentiment Signal** stronger. The **Entity Analysis** continued to behave mainly as a supporting signal, while the **Theme Analysis** remained the strongest standalone signal throughout the project. The combined **Earnings and Theme Signal** also continued to explain company-level behaviour better than the overall market behaviour, even after the additional data was introduced.

Another important understanding emerged while interpreting both projects together. Some of the analyses were simply too complex to be understood through visualisations alone. Although the charts made the comparisons much easier, they could not explain the complete behaviour of the reusable systems. Because of this, detailed interpretation became necessary to convert the numerical and textual findings into meaningful analytical understanding. Instead of simply describing what had changed, the interpretation explained why those changes occurred and what they meant for the behaviour of both analytical systems after the append process.

Overall, the interpretation showed that both reusable systems continued to remain stable after the new data was added. More importantly, the original understanding of both projects also remained largely unchanged. The Machine Learning Project continued to produce stable and reliable results, while the Natural Language Processing Project continued to show that the **Theme Analysis** remained the strongest NLP signal for understanding market behaviour. This showed that the reusable systems were not only capable of processing additional data successfully but were also able to preserve the original analytical understanding as the data evolved over time.

So, the answer to this major question is that interpretation is much more than simply explaining the visuals. It converts both the numerical and textual findings into meaningful analytical understanding. I think this became one of the most important stages of the entire project because it connected the technical results with the practical understanding that emerged from them. It also confirmed that the Machine Learning and Natural Language Processing Projects remained reusable after adding the new data and that the reusable analytical framework continued to provide stable, meaningful, and reliable understanding whenever the workflow was executed again.

Fourth Major Question of the Project -

After completing **Part A** and **Part B** of the Word document, which focus on the **Machine Learning** and **Natural Language Processing** projects individually, the final major question is about understanding them together as one complete reusable analytical framework. Until now, both projects have been analysed independently. However, after completing the complete series of four projects, starting with the Machine Learning Project, followed by the Natural Language Processing Project, then the Reusable Systems Project, and finally the Reusable Analytical Framework Validation Project, it becomes important to understand the complete framework instead of looking at each project separately.

The fourth major question is therefore about understanding how both the ML and NLP projects work together, where they provide common understanding, where they differ, and how each project contributes towards the complete reusable analytical framework. It also focuses on understanding whether combining both projects provides a better analytical understanding than studying them individually, what practical understanding emerges from the complete framework, and what the overall contribution of the entire project series is.

This final understanding should explain not only what each individual project achieved, but also what I learned after completing all four projects together. It should provide the overall understanding of the reusable analytical framework, explain its future usefulness and practical applications, and show the overall analytical contribution of the complete project series.

Finally, after completing the analytical understanding of the complete framework, I also want to reflect on my journey throughout these four projects. The purpose of this reflection is not to discuss the technical results again, but to explain how my understanding evolved during the project. This journey actually began with my initial attempt to learn Artificial Intelligence through **Google Vertex AI**, which was originally my first approach but did not work as I expected. That challenge eventually led me towards **BigQuery SQL**, where I discovered the concept of SQL Views. Looking back now, I realise that this completely changed the direction of the project. From there, the journey gradually evolved into the Machine Learning Project, the Natural Language Processing Project, the Reusable Systems Project, and finally the Reusable Analytical Framework Validation Project. Looking back at this journey will help explain not only what I built, but also how my thinking, analytical understanding, and overall approach towards Machine Learning, Natural Language Processing, and reusable analytical systems developed

throughout the complete series of projects.

Sub Question 1 -

After understanding the updated results of both the **Machine Learning Project** and the **Natural Language Processing Project** separately, the next question is whether there are some common understandings that emerge when both projects are viewed together. Even though both projects work on completely different types of data, use different analytical methods, and follow different workflows, they were both created with the same overall objective of building reusable analytical systems. Because of this, it becomes important to understand whether they reached any common conclusions after the new data was added.

The first thing that becomes noticeable is that both projects continued to work successfully after the append process. The internal implementation of both projects is completely different. The Machine Learning Project works mainly through **BigQuery ML models** and downstream SQL views, whereas the Natural Language Processing Project depends on **Python-based NLP extraction** before the analysis continues inside BigQuery SQL. Even with these architectural differences, both projects were able to process the newly added data without changing the overall workflow.

This immediately gives one important understanding. Reusability does not depend on using the same analytical method. A reusable system can be built in different ways, provided the overall workflow remains capable of accepting and processing future compatible data without requiring the complete analytical structure to be rebuilt.

However, while thinking about this further, another question becomes more important. Is reusability alone enough? If a project continues processing new data but produces completely different analytical conclusions every time, then the practical usefulness of that reusable structure becomes questionable. A reusable analytical system should not only continue working technically, but should also continue producing stable analytical understanding over time.

When I compare both projects from this perspective, I notice that this stability is present in both of them. In the **Machine Learning Project**, the regression model remained stable, the classification model continued producing almost identical results, and the Reliability Analysis again identified the Logistic model as the most stable model after the new market data was added. Similarly, in the **Natural Language Processing Project**, many numerical values changed because a much larger amount of financial news became available, but the overall interpretation remained the same. Theme Analysis continued to remain the strongest signal, while Sentiment Analysis and Entity Analysis continued behaving mainly as supporting signals.

This makes me realise that the real success of a reusable analytical system is not simply that it accepts new data, but that the analytical understanding itself remains stable while adapting to that new information.

Another common observation relates to the amount of data. Initially, I expected that adding more data might naturally improve the overall analytical results. However, the updated results do not fully support that expectation.

In the **Machine Learning Project**, adding six months of additional market data did not significantly improve the models. The regression errors increased only slightly, while the classification models remained almost identical to their earlier performance. Similarly, in the **Natural Language Processing Project**, adding a much larger amount of financial news increased the number of observations considerably, but it did not automatically strengthen every NLP signal. Sentiment Analysis remained close to the same level, the combined signal improved only marginally, and Theme Analysis continued to provide the strongest understanding.

This suggests that the quality and nature of the information entering the system often matter much more than simply increasing the quantity of data.

One more common understanding appears while comparing both projects. Increasing analytical complexity did not necessarily produce a better understanding of the market.

During the **Machine Learning Project**, increasing model complexity through ensemble methods improved some performance measures, but the Reliability Analysis continued to show that the Logistic model remained the most stable model when evaluated on unseen data. Similarly, in the **Natural Language Processing Project**, combining multiple NLP signals did not create a stronger signal than Theme Analysis by itself. Instead, Theme Analysis continued to provide the clearest understanding because it focused on the overall context and meaning of the financial news rather than isolated words or individual entities.

This shows that a more complex analytical approach does not automatically provide a more meaningful understanding. In many situations, a simpler analytical structure can produce results that are easier to interpret and more reliable over time.

Another reassuring observation also becomes visible when both projects are viewed together. Although many numerical values changed after the append process because the datasets became larger, the original conclusions remained largely unchanged. The additional data mainly strengthened confidence that the earlier findings were not accidental rather than replacing them with entirely new conclusions.

Finally, perhaps the biggest common understanding from both projects is related to their overall objective. When I first started these projects, prediction naturally appeared to be one of the main goals. However, after completing all four projects and analysing the updated results, I realised that both projects gradually shifted towards something more important than prediction itself.

The **Machine Learning Project** helped explain how the market behaves by analysing stable numerical relationships, while the **Natural Language Processing Project** helped explain why the market behaves by analysing financial news, themes, entities, and contextual information. Together, they became systems for understanding market behaviour rather than systems designed only to maximise predictive performance.

Overall, this comparison shows that although the Machine Learning Project and the Natural Language Processing Project were built using different data, different workflows, and different analytical methods, they ultimately reached several common understandings. Both demonstrated that a reusable analytical system should preserve its analytical understanding as new data is added, that more data does not automatically improve analytical quality, and that increasing complexity does not necessarily produce better understanding. These common conclusions provide the foundation for understanding the complete reusable analytical framework as a whole.

Sub Question 2 -

After understanding the common learnings that emerged from both the **Machine Learning Project** and the **Natural Language Processing Project**, another important question comes into my mind. If both projects reached many similar conclusions about reusability and stability, then what actually makes them different? Although both projects are part of the same reusable analytical framework, they were never built to solve exactly the same problem. Instead, they were designed to understand different aspects of the financial market, and this difference becomes much clearer when they are viewed together.

The first difference becomes visible in the foundation on which both projects were built. The **Machine Learning Project** started with structured numerical data such as stock prices, returns, and market movements. The complete analysis depended on historical numerical relationships and whether those relationships remained stable after the new data was added. The **Natural Language Processing Project**, on the other hand, started with unstructured financial news. Before any meaningful analysis could begin, the news first had to be converted into structured information using different NLP techniques. This itself shows that both projects begin from completely different foundations even though they ultimately become part of the same reusable analytical framework.

As I compare both projects further, I also realise that they explain different dimensions of the financial market. The **Machine Learning Project** mainly explains how the market behaves by analysing numerical relationships and checking whether those relationships remain stable over time. The **Natural Language Processing Project** explains what is happening in the market by analysing financial news, themes, entities, and the overall meaning of the available information. This shows that both projects answer different questions about the same market rather than trying to produce the same type of understanding.

Another difference becomes visible after adding the new data. In the **Machine Learning Project**, six months of additional market data produced only small changes in the behaviour of the models. Most of the regression and classification results remained stable, showing that the trained models continued to generalise well on the larger unseen dataset. The **Natural Language Processing Project** behaved differently because the additional financial news introduced many new entities, themes, and market events. As a result, many individual NLP values changed after the append process. However, despite these changes, the overall interpretation of the project remained largely unchanged. This makes me realise that numerical market data usually changes gradually, whereas financial news changes continuously because new events keep entering the market.

While comparing both projects, I also notice that their most important findings emerged from different areas. In the **Machine Learning Project**, the **Reliability Analysis** became the most important finding because it explained something that the traditional evaluation metrics could not show. It demonstrated that the **Logistic Classification** model remained the most stable model after moving from the training data to unseen observations. In the **Natural Language Processing Project**, the **Theme Analysis** became the strongest finding because it provided a much deeper understanding of market behaviour than either **Sentiment Analysis** or **Entity Analysis**. Rather than focusing on individual words or entities, Theme Analysis explained the broader meaning and context behind the financial news.

Another important difference appears in the challenges faced by both projects. The **Machine Learning Project** mainly focused on identifying which model remained the most stable after adding new data because the traditional evaluation metrics showed only small differences between the models. The Reliability Analysis provided the additional understanding needed to distinguish them. In contrast, the **Natural Language Processing Project** faced the challenge of converting unstructured financial news into meaningful analytical signals. Throughout the project, it gradually became clear that understanding the overall theme of the news was much more useful than analysing only individual words or entities. Although both projects solved different analytical problems,

they both required deeper investigation before meaningful conclusions could be reached.

One more difference also becomes visible while looking at the reusable workflow itself. The **Machine Learning Project** followed a highly structured workflow where most of the processing was performed directly inside **BigQuery**, and the trained models automatically generated updated results after new data was appended. The **Natural Language Processing Project**, however, followed a more modular workflow because **Python** was required to process the financial news before the extracted information could be analysed further inside **BigQuery**. Theme Analysis also required continuous manual interpretation and maintenance before the final themes could be prepared. This shows that the reusable architecture itself depends on the nature of the data being analysed.

Overall, comparing both projects makes one thing very clear. Although both projects successfully demonstrate reusable analytical systems, they were designed to explain different aspects of the financial market. The **Machine Learning Project** focuses on historical numerical behaviour, model stability, and prediction consistency, while the **Natural Language Processing Project** focuses on financial information, market context, and the meaning behind the news. Rather than competing with each other, both projects complement each other by explaining different dimensions of the same market. This understanding naturally prepares the next stage of the project, where both approaches can be brought together to understand how they complement each other in practical analytical applications

Sub Question 3 -

After understanding the common and different learnings that emerged from both the **Machine Learning Project** and the **Natural Language Processing Project**, another important question comes into my mind. Both projects have now successfully demonstrated that they remain reusable after adding the new data, but what practical value do they actually provide? It becomes important to understand whether this complete reusable analytical framework can be useful beyond this project and how it can contribute to practical financial analysis.

One thing becomes very clear while looking at both projects together. Throughout the complete series of projects, the objective was never to build a system that predicts the financial market with complete accuracy. Instead, the objective was to build a reusable analytical framework that continues to understand new financial data whenever it is added. The append analysis also supports this understanding because both projects

continued producing stable analytical results without changing their overall conclusions after the new data was introduced.

As I compare both projects from a practical perspective, I realise that they should be used primarily for understanding the market rather than directly predicting it. The **Machine Learning Project** showed that the models remained stable after adding the new data, but it also demonstrated that higher model performance does not always lead to better long-term stability. The **Reliability Analysis** became one of the most important findings because it explained how the models behaved on unseen data instead of simply comparing evaluation metrics. Similarly, the **Natural Language Processing Project** showed that **Theme Analysis** explained market behaviour much better than **Sentiment Analysis** or **Entity Analysis**, but it also showed that financial news alone cannot explain every market movement. This makes me realise that the practical value of the framework lies more in improving the understanding of market behaviour than in prediction alone.

Another practical understanding is that both projects explain different parts of the same financial market and therefore work much better together than individually. The **Machine Learning Project** explains how the market behaves by analysing historical numerical relationships and checking whether those relationships remain stable after adding new data. The **Natural Language Processing Project** explains what information is influencing the market by analysing financial news, themes, entities, and the overall market context. When both projects are viewed together, they provide a broader understanding because one explains the numerical behaviour while the other explains the information behind that behaviour.

While working through the append analysis, I also understood that this framework should be viewed as a **decision-support system** rather than a **decision-making system**. Many important conclusions during the project required analytical interpretation before the final understanding could be reached. The **Reliability Analysis** was required to identify the most stable Machine Learning model, while **Theme Analysis** was required to identify the strongest NLP signal. Simply looking at the numerical outputs would not have produced the same understanding. This shows that the framework supports financial analysis by providing structured analytical understanding, but the final interpretation should still depend on human judgement and financial knowledge.

Another practical benefit becomes visible through the reusable workflow itself. One of the main objectives of this project was to understand whether the analytical process could continue working without rebuilding the complete framework whenever new data became available. The append analysis showed that the **Machine Learning Project** continued generating updated analytical results without retraining the models, while the **Natural Language Processing Project** continued producing updated signals using the same reusable workflow after new financial news was added. This demonstrates that the

framework can continue supporting future financial analysis without redesigning the complete analytical process every time the data changes.

I also realise that the framework is flexible enough to support different types of financial analysis. The **Machine Learning Project** can be used for understanding historical market behaviour, evaluating model stability, and analysing numerical relationships over time. The **Natural Language Processing Project** can be used for understanding financial news, identifying important market themes, analysing changing financial events, and explaining the information available during different market situations. Together, both projects provide a broader analytical understanding that can support equity research, market monitoring, investment research, financial reporting, and academic financial studies.

Finally, one of the biggest practical understandings from this complete framework is that its real value does not come from automation alone. Many systems can automatically process new data, but if the analytical understanding changes completely every time the data changes, then the practical usefulness of that system becomes very limited. This project showed that although many numerical values and individual signals changed after the append process, the overall understanding of both the **Machine Learning Project** and the **Natural Language Processing Project** remained stable. This makes analytical reusability much more valuable than automation by itself.

Overall, this practical analysis shows that the reusable framework should not be viewed as a system that predicts the financial market with complete certainty. Instead, it should be viewed as a reusable analytical framework that continues to generate stable and meaningful financial understanding as new structured and unstructured data becomes available. This practical understanding naturally prepares the final stage of the project, where the complete journey, overall learnings, and the final conclusions from all four projects can be brought together.

Sub Question 4 -

After understanding the common learnings, the different learnings, and the practical understanding that emerged from both the **Machine Learning Project** and the **Natural Language Processing Project**, one final question comes into my mind. Till now, I have mostly discussed every project separately. Even though the **Reusable Systems Project** connected both of them, I have never really looked at the complete work as one analytical framework. Since this is the final project of the complete series, it becomes important to understand what all four projects together finally contribute and what this complete reusable analytical framework actually becomes.

As I look back at the complete journey, I realise that this framework was never built in one stage. Every project answered a different question and gradually became the foundation for the next project. The **Machine Learning Project** helped me understand the numerical behaviour of the financial market using structured data. After that, the **Natural Language Processing Project** expanded that understanding by analysing financial news, themes, entities, and market context. The **Reusable Systems Project** then connected both projects into one reusable workflow, making it possible to process new data without rebuilding the complete analytical process. Finally, the **Reusable Analytical Framework Validation Project** verified whether the complete framework continued to produce stable analytical understanding after new data was added. Looking back now, I realise that none of these projects was independent. Each one became another step towards building the complete analytical framework.

Another important understanding also becomes visible when I stop looking at the projects individually. If I study only the **Machine Learning Project**, it explains the numerical behaviour of the market. Similarly, if I study only the **Natural Language Processing Project**, it mainly explains the information and context behind that market behaviour. However, when both projects are viewed together inside the reusable framework, they provide a much broader understanding of the financial market. The numerical relationships, financial information, and reusable workflow all become part of one complete analytical system instead of remaining separate projects.

While thinking about the complete journey, I also realise that my own understanding gradually changed throughout these projects. In the beginning, my focus was mainly on understanding **Machine Learning models** and **Natural Language Processing signals**. However, after completing the **Reusable Systems Project** and the append analysis, reusability itself became the central understanding of the complete framework. Building models and extracting signals remained important, but a much bigger question gradually emerged: **Can the same analytical workflow continue producing meaningful understanding whenever new data is added?** The append analysis answered that question by showing that both projects remained stable even though the datasets became larger and many observations changed.

Another important understanding is that the complete framework demonstrated stability in two different ways. The reusable workflow continued processing both structured and unstructured data without changing its overall architecture, demonstrating **operational stability**. At the same time, the analytical understanding of both the **Machine Learning Project** and the **Natural Language Processing Project** also remained stable after the append process, demonstrating **analytical stability**. This makes me realise that the framework remained successful not only because it continued to execute correctly, but also because it continued to preserve its original analytical understanding as the data evolved.

I also understand that this framework explains the financial market from multiple perspectives instead of depending on only one analytical approach. The **Machine Learning Project** explains the historical numerical behaviour of the market, while the **Natural Language Processing Project** explains the financial information and market context available during different situations. Together, both projects provide a broader understanding than either structured numerical data or financial news could provide independently.

Finally, one of the biggest understandings from this complete journey is that this framework should continue to evolve. New market data and new financial news will continue to become available in the future. The reusable workflow allows the same analytical process to continue working without rebuilding the complete framework, while the analytical conclusions can continue to be validated whenever additional data is added. This makes the framework dynamic rather than static, allowing it to preserve its original analytical structure while continuously learning from new information.

Overall, after completing all four projects, one understanding becomes very clear. The objective of this complete work gradually evolved from understanding individual **Machine Learning models** and **Natural Language Processing techniques** to building one reusable analytical framework capable of understanding both structured numerical data and unstructured financial information together. Rather than focusing only on prediction or automation, the complete framework focuses on preserving meaningful analytical understanding as the data continues to evolve. Looking back at the complete journey, I think that became the biggest learning from all four projects. This understanding naturally prepares the answer to the final major question of the project, where the complete journey, the overall learning, and the overall contribution of the entire project series can be brought together as one conclusion.

Answer of the fourth major question -

After completing the **Machine Learning Project**, the **Natural Language Processing Project**, the **Reusable Systems Project**, and finally the **Reusable Analytical Framework Validation Project**, one overall understanding became very clear. Although each project was developed independently and answered a different analytical question, they were never intended to remain separate. Instead, each project gradually became the foundation for the next one, and together they formed one complete reusable analytical framework for understanding financial markets.

The **Machine Learning Project** provided the first stage of this framework by analysing structured numerical data such as stock prices, returns, and market movements. It explained how different Machine Learning models behaved, how stable they remained

after adding new data, and how the **Reliability Analysis** became a better way of understanding long-term model behaviour than relying only on traditional evaluation metrics. This project established the numerical foundation of the complete framework.

The **Natural Language Processing Project** expanded this understanding by introducing unstructured financial news into the analysis. Instead of analysing numerical relationships, it focused on understanding financial information through **Sentiment Analysis, Entity Analysis, Theme Analysis**, and different combined signals. Throughout the project, **Theme Analysis** consistently emerged as the strongest standalone NLP signal because it explained the broader meaning of financial news rather than analysing only individual words or entities. This project added the informational and contextual understanding that numerical analysis alone could not provide.

After completing both independent projects, the **Reusable Systems Project** connected them into one reusable analytical workflow. Rather than rebuilding the complete analysis whenever new data became available, the framework allowed the same analytical process to continue generating updated results using both structured and unstructured data. At this stage, the focus of the work gradually shifted from building individual analytical models to building one reusable analytical system.

The final append analysis then became the validation stage of the complete framework. New market data and financial news were added to both projects without changing their original workflow. Although many numerical values, themes, entities, and observations changed after the append process, the overall analytical understanding remained stable. The **Machine Learning Project** continued to identify the **Logistic Classification** model as the most stable model through the **Reliability Analysis**, while the **Natural Language Processing Project** continued to identify **Theme Analysis** as the strongest standalone NLP signal. This demonstrated that the reusable analytical framework remained both operationally and analytically stable after adding new data.

Another important understanding that emerged after completing all four projects is that the **Machine Learning Project** and the **Natural Language Processing Project** complement each other rather than compete with each other. The Machine Learning Project explains **how** the financial market behaves by analysing historical numerical relationships, while the Natural Language Processing Project explains **what** information and market context influence that behaviour. Together, they provide a broader understanding of the financial market than either project could provide independently.

The complete framework also showed that reusability alone is not enough. A reusable system becomes meaningful only when it continues to produce stable analytical understanding after new data is added. Throughout the append analysis, both projects demonstrated this behaviour. Even though the datasets became larger and many

individual observations changed, the overall conclusions remained consistent. This showed that preserving analytical stability is just as important as building a reusable workflow.

Another important learning from the complete framework is that increasing analytical complexity does not always improve understanding. In the **Machine Learning Project**, the Ensemble models produced stronger evaluation metrics, but the **Reliability Analysis** identified the **Logistic Classification** model as the most stable model when evaluated on unseen data. Similarly, in the **Natural Language Processing Project**, combining multiple NLP signals did not produce a stronger analytical signal than **Theme Analysis** alone. These findings showed that a simpler analytical approach can sometimes provide a more reliable understanding than a more complex one.

The complete framework also demonstrated that its practical value lies in supporting financial analysis rather than replacing human decision-making. Throughout the project, many important conclusions required analytical interpretation before meaningful understanding could be developed. The framework therefore acts as a decision-support system that helps explain market behaviour using both structured numerical analysis and unstructured financial information, while the final interpretation continues to depend on human judgement and financial knowledge.

Looking back at the complete journey, I also realise that the objective of this work gradually evolved over time. It did not remain limited to building Machine Learning models, analysing financial news, or automating analytical workflows individually. Instead, it gradually became an attempt to build one reusable analytical framework that can continue to understand both structured numerical data and unstructured financial information while preserving its analytical understanding as new data continues to become available.

Therefore, the overall contribution of this complete series of projects is not simply the development of individual Machine Learning or Natural Language Processing models. Its real contribution is the development and successful validation of a reusable analytical framework that combines structured numerical analysis, unstructured financial information, reusable workflows, and analytical stability into one complete system. Looking back now, I think that became the biggest outcome of the entire journey. More than building models or processing data, I learned how different analytical approaches can work together to produce a deeper understanding of financial markets. That, more than any individual result, is what I will carry forward from these four projects.

Answer to the Problem Statement

After completing the complete series of four projects, I can now say that the original problem statement has been successfully answered. The objective of this project was never to build new Machine Learning models or develop new Natural Language Processing techniques. Instead, the objective was to understand whether the existing analytical structure could be converted into a reusable analytical framework that continues to work correctly whenever new compatible data is added with minimum manual effort.

Looking back at the complete work, the **Machine Learning Project** successfully demonstrated this behaviour. The analytical workflow was built using BigQuery SQL views, allowing the same reusable structure to process updated market data without changing the overall workflow. After adding six months of new market data, the framework generated updated regression, classification, and Reliability Analysis results without retraining the models. Although some numerical values changed slightly, the overall analytical understanding remained the same. The Logistic Classification model continued to remain the most stable model, showing that the reusable ML workflow preserved both its operational continuity and its analytical understanding after the append process.

The **Natural Language Processing Project** presented a different challenge because BigQuery does not directly support NLP functions in the same way that it supports Machine Learning. To overcome this limitation, Python was used as an external processing layer to extract the NLP signals before transferring the processed information back into BigQuery for further analysis. Although this workflow required an additional processing stage and more manual decisions, the reusable structure successfully generated updated Sentiment, Entity, Theme, and combined analytical signals after the new financial news was added. The append analysis also showed that **Theme Analysis** continued to remain the strongest standalone NLP signal, while the overall analytical understanding of the project remained unchanged even after introducing a much larger news dataset.

Another important objective of this project was to reduce repetitive manual work while keeping the analytical process simple, reusable, and manageable. Looking back at the complete workflow, I think this objective was also achieved. The reusable structures developed throughout the projects allowed new structured and unstructured data to move through the same analytical process without rebuilding the complete project. The use of SQL views, modular processing, and view-based analytical checkpoints made it possible to update the framework with minimum structural changes while preserving the original analytical logic.

The project also showed that reusability alone is not enough. A reusable analytical system becomes meaningful only when it continues to produce stable analytical understanding after new data is added. Throughout the append analysis, both the Machine Learning and Natural Language Processing Projects demonstrated this behaviour. Although many numerical values, entities, themes, and observations changed after the append process, the original conclusions of both projects remained valid. This showed that the framework remained not only operationally reusable but also analytically stable.

Another important understanding that emerged during the project is that the real value of the framework does not come from automation alone. The reusable workflow certainly reduced repetitive manual work, but its biggest contribution was preserving analytical understanding as the data continued to evolve. Rather than rebuilding models or redesigning workflows whenever new data became available, the same analytical process continued to generate updated results while preserving the original understanding developed during the earlier projects.

Looking back at the complete journey, I also realise that the objective gradually evolved beyond what I had originally planned. What started as an attempt to understand Machine Learning and Natural Language Processing separately eventually became an attempt to build one reusable analytical framework capable of working with both structured numerical data and unstructured financial information. The append analysis became the final validation that this framework could continue operating while preserving its analytical understanding over time.

Overall, I believe the original problem statement has been successfully answered. The complete project demonstrates that both the Machine Learning and Natural Language Processing workflows can be integrated into one reusable analytical framework that continues to process new data, preserve its original analytical understanding, and generate updated results with minimum manual effort. More importantly, it shows that the real value of a reusable analytical system lies not simply in processing new data automatically, but in continuing to provide stable, meaningful, and reliable financial understanding as the data evolves. That, in the end, is the biggest outcome of this complete series of four projects.

Summary of the Project

This project is the final project in the complete series of four projects, and its main purpose is not to build another Machine Learning model or another Natural Language Processing technique. Instead, it focuses on understanding the overall work that has already been done and validating whether the complete reusable analytical framework continues to behave in the same way after new compatible data is added.

The complete journey started with the Machine Learning Project, where I tried to understand whether machine learning models could explain the relationship between stock returns and market returns. After that, I moved towards the Natural Language Processing Project because numerical market movements alone cannot explain why the market behaves in a particular way. Financial news, themes, entities, and other textual information also influence market behaviour, so the NLP Project became the second part of the overall analytical framework.

While completing these two projects, I realised that I did not want them to remain one-time analyses. Since both structured market data and financial news continue to change over time, I wanted to build a reusable workflow that could process future data without rebuilding the complete analytical structure again. This understanding eventually led to the Reusable Systems Project, where the overall architecture of both projects was organised into one reusable analytical workflow.

However, even after completing the reusable system, one important thing was still missing. I had created a framework that could process new data, but I had not actually validated whether the original analytical understanding remained the same after new compatible data entered the system. Therefore, this project mainly focuses on validating the complete reusable framework by adding six months of new structured market data and financial news, comparing the updated results with the original findings, and understanding whether the overall behaviour of the framework remained stable.

Another important objective of this project is to organise the complete findings in a much better way. The original and updated results have been organised into structured comparison tables, visualisations, and detailed interpretations so that both the Machine Learning and Natural Language Processing Projects can be understood together instead of separately. This also helps in understanding not only what changed after adding the new data but also what remained unchanged, which findings became stronger, which findings remained weak, and what limitations still continue to exist.

While working on this project, I also realised that the purpose of a reusable analytical framework is not simply to automate future analysis. Automation alone does not make a system useful. A reusable system becomes valuable only if it continues to produce

stable, meaningful, and reliable analytical understanding whenever new compatible data is added. Therefore, the main focus of this project gradually shifted from simply updating the results towards validating the stability of the complete analytical framework itself.

Overall, this project provides the closure for the complete four-project journey. It validates the reusable analytical framework, compares the original and updated findings, organises the complete analytical understanding into a structured form, and explains how both the Machine Learning and Natural Language Processing Projects work together as one reusable analytical system. More importantly, it also establishes a foundation that can continue to be reused whenever new data becomes available in the future, while preserving the original analytical understanding developed throughout this complete series of projects.

Key Findings

One of the biggest findings from this project is that the reusable analytical framework remained stable even after adding new compatible data. The purpose of adding the additional six months of market data and financial news was not simply to generate updated outputs, but to understand whether the original analytical findings continued to remain valid. After comparing the original and updated results, I found that most of the important conclusions from both the Machine Learning and Natural Language Processing Projects remained unchanged, which increases confidence in the overall framework.

Another important finding is that the Machine Learning Project behaved much more consistently than I originally expected. Even after increasing the size of the test dataset, the regression results remained almost the same, the classification models continued to produce very similar performance, and the Reliability Analysis again identified the Logistic Classification model as the most stable model. This showed that the trained models generalised well on the newly added unseen data and that the reusable workflow continued to behave as expected.

The Natural Language Processing Project produced a different type of understanding. Unlike the Machine Learning Project, where most of the findings remained numerically stable, the NLP Project became richer after the addition of the new financial news. More entities, themes, and observations became available, which increased the amount of analytical understanding. At the same time, it also increased the amount of manual work required to maintain the Theme Analysis and other NLP signals. This showed that adding more data improves the richness of the analysis, but it also increases the complexity of maintaining the overall NLP workflow.

One important observation throughout the project is that Theme Analysis continued to remain the strongest standalone NLP signal. Even though many individual themes changed after the addition of the new data, the overall understanding remained the same. Similarly, the Entity Analysis became richer because of the larger amount of financial news, while the Sentiment Analysis continued to remain a weaker signal even after increasing the number of observations. This helped me understand that not every analytical signal improves simply by increasing the amount of available data.

Another important finding is that increasing analytical complexity does not always improve analytical understanding. In the Machine Learning Project, the more complex ensemble models produced better training performance but did not improve the overall stability identified through the Reliability Analysis. Similarly, in the NLP Project, combining multiple signals did not produce a stronger understanding than Theme Analysis alone. This showed that a simpler and more stable analytical approach can sometimes be more valuable than a more complex one.

One thing I also realised while completing this project is that reusability alone is not enough. A reusable workflow becomes meaningful only if the analytical understanding also remains stable after new data is added. Throughout this project, both the Machine Learning and Natural Language Processing workflows continued to process the new data successfully, while the overall conclusions remained largely unchanged. This showed that the framework achieved not only operational reusability but also analytical stability.

Overall, the biggest finding from this complete project is that the reusable analytical framework continued to preserve its original understanding even after the append process. The framework successfully processed new structured and unstructured financial data, validated the original findings, highlighted the strengths and limitations of both projects, and demonstrated that meaningful analytical understanding can be preserved while the data continues to evolve. This became the most important outcome of the complete four-project journey.

What Worked

One of the things that worked really well throughout this project was the reusable workflow that had already been developed in the previous projects. Since the overall architecture was already organised using BigQuery SQL views, most of the Machine Learning workflow continued to operate without requiring any structural changes after the new market data was added. The updated results were generated through the same downstream analytical flow, which showed that the reusable architecture behaved as expected.

Another thing that worked well was the stability of the Machine Learning models themselves. Even after adding six months of additional market data, the overall findings remained almost identical to the original project. The regression results showed only marginal changes, the classification models continued to perform at a similar level, and the Reliability Analysis again identified the Logistic Classification model as the most stable model. This strengthened my confidence that the original modelling approach was not limited to one particular dataset but continued to remain useful after new compatible data was introduced.

The comparison between the original and updated results also worked much better than I initially expected. Organising both sets of results into structured comparison tables made it much easier to understand what had actually changed and what had remained the same. Instead of analysing the updated outputs independently, comparing them directly with the original findings provided a much clearer understanding of the behaviour of the reusable framework.

The use of MS Excel for organising the comparison tables and visualisations also proved to be a practical decision. Initially, I planned to use Power BI because it connects directly with BigQuery SQL. However, after completing the previous projects, I realised that BigQuery only contained the updated results, whereas the original findings were stored separately in the Output Fact Sheets. Excel allowed both the original and updated results to be organised together in one place, making the comparison and visualisation process much easier.

Another thing that worked well was separating the evidence from the interpretation. The Excel workbook became the main evidence file containing the comparison tables, visualisations, and short interpretation boxes, while the separate Word document focused on the detailed analytical interpretation of the results. This separation made the overall project much easier to understand because the evidence and the detailed explanation no longer competed for the same space.

The overall structure of the project also proved to be effective. Dividing the work into major questions and sub-questions made the complete analysis much easier to organise. Each question focused on solving one specific problem before moving to the next one, which helped in building the final understanding gradually instead of trying to explain everything at once. Looking back at the complete project, this problem-and-answer structure made the overall analytical journey much more organised and easier to follow.

One thing that worked particularly well throughout the complete series of projects was the idea of validating the original findings instead of trying to create completely new ones. The append analysis showed that the original conclusions of both the Machine Learning

and Natural Language Processing Projects largely remained valid after the new data was added. This shifted the focus from generating new outputs towards validating existing analytical understanding, which ultimately became one of the strongest aspects of the complete reusable analytical framework.

Overall, the biggest thing that worked throughout this project was not any individual model, signal, or visualisation. It was the overall reusable analytical framework itself. The framework successfully accepted new compatible data, generated updated results, preserved the original analytical understanding, and provided a structured way to compare, validate, and interpret those findings. Looking back at the complete project, this became the strongest evidence that the overall analytical framework remained both operationally reusable and analytically stable.

What Did Not Work

One of the biggest things that did not work as I originally expected was the Natural Language Processing workflow. While the Machine Learning Project behaved almost exactly as I had planned, the NLP Project required much more manual work than I had initially anticipated. Since BigQuery does not directly support NLP functions, an external Python workflow had to be created before the processed results could be brought back into BigQuery. This made the overall workflow less smooth than the Machine Learning workflow and increased the amount of manual effort required whenever new data was added.

Another thing that did not work as expected was the assumption that adding more data would automatically improve the analytical results. Before adding the new six months of financial news, I expected that the larger dataset would strengthen most of the NLP signals. However, that did not happen. Although the amount of available information increased considerably, the Sentiment Analysis continued to produce almost the same level of consistency as before. Similarly, increasing the number of observations did not automatically improve the overall predictive quality of the combined signals. This helped me understand that the quality of the analytical signal depends much more on the way it is constructed than simply on the amount of available data.

The Theme Analysis also presented an unexpected challenge. Even though it remained the strongest NLP signal throughout the project, it required continuous manual maintenance whenever new financial news was added. New phrases and new market events had to be manually mapped into existing themes or completely new themes had to be created. Although this maintenance improved the quality of the analysis, it also showed that the Theme Analysis cannot currently operate as a fully automated process.

The Entity Analysis also did not perform as cleanly as I initially expected. After adding the new financial news, the number of entities increased significantly, but many entities were duplicated because of different naming conventions, while several high-frequency entities contributed very little analytical understanding. This introduced additional noise into the analysis and showed that the entity extraction process still requires further refinement before it can become a stronger standalone analytical signal.

Another thing that did not work particularly well was the current multi-signal approach used in the NLP Project. I expected that combining Sentiment, Entity, and Theme signals would produce a stronger overall analytical signal. However, the combined signal did not improve as much as expected. Even after adding the new data, the overall consistency remained relatively weak. Looking back, I think the limitation was not in the individual signals themselves but in the simple way they were combined. This suggests that the signal combination logic itself requires further improvement in future work.

I also realised that some results were difficult to communicate effectively through visualisations alone. Although charts made numerical comparisons easier to understand, they were not sufficient for explaining complex analytical findings such as the Reliability Analysis in the Machine Learning Project or the Theme Analysis in the Natural Language Processing Project. Because of this, additional interpretation documents became necessary to explain the actual meaning behind those results. This showed that visualisation alone cannot always communicate analytical understanding effectively.

Finally, I also realised that building a reusable analytical framework does not completely eliminate manual work. The overall workflow became much more reusable than before, but certain parts of the process, especially within the NLP Project, still depend on analytical judgement, manual theme updates, and external processing. Therefore, the framework successfully reduced repetitive work, but it did not completely automate every stage of the analytical process.

Overall, the things that did not work were not failures of the project itself. Instead, they became some of the most important learnings throughout the complete journey. They highlighted where the current framework still requires refinement, where additional automation can be introduced in the future, and where analytical judgement will probably continue to remain necessary even as the reusable framework evolves.

Final Takeaway

Looking back at the complete journey, I realise that this series of projects gradually became much bigger than what I had originally planned. In the beginning, my objective was simply to understand Machine Learning and Natural Language Processing in the

context of financial analysis. However, as each project was completed, the focus slowly shifted from learning individual techniques to understanding how different analytical methods can work together inside one reusable framework.

One of the biggest takeaways from this complete journey is that financial analysis is not only about building better models or creating more complex algorithms. It is about developing analytical understanding that continues to remain useful even when new data becomes available. Throughout the append analysis, many numerical values changed, new financial news entered the system, and additional observations became available, but the overall understanding of both the Machine Learning and Natural Language Processing Projects remained largely unchanged. This showed me that preserving analytical stability is much more important than simply generating new results every time the data changes.

Another important understanding I gained is that different analytical approaches should not be viewed as competing with one another. The Machine Learning Project and the Natural Language Processing Project solve different parts of the same problem. One explains the numerical behaviour of the market, while the other explains the information and context surrounding that behaviour. Together, they provide a much broader understanding than either approach could provide individually. This became one of the strongest lessons from the complete framework.

I also realised that reusability is not achieved simply by automating a workflow. A reusable analytical system becomes valuable only when it continues to produce meaningful and reliable understanding after new compatible data is added. Throughout this project, the reusable framework continued to process both structured and unstructured data while preserving its original analytical understanding. This showed that the real value of the framework lies not only in reducing manual work but also in maintaining consistency as the analytical process evolves over time.

Finally, this complete journey changed the way I think about analytical projects themselves. Earlier, I used to see Machine Learning, Natural Language Processing, SQL, Python, and other tools as separate technologies that needed to be learned individually. After completing these four projects, I now see them as different parts of one analytical ecosystem, where each tool contributes to solving a different part of the overall problem. The real objective is no longer to use more tools, but to understand when each tool should be used and how they can work together to produce meaningful analytical understanding.

Overall, this project provides the closure to the complete four-project journey. More importantly, it provides a reusable analytical foundation that can continue to evolve as new financial data becomes available. Although the framework will continue to grow and

improve in the future, the overall understanding developed throughout these four projects will remain the foundation on which those future improvements can be built. Looking back at the complete journey, I believe this has been the most valuable outcome of the entire project series.

Project Journey and Learning Reflection

I still can't believe that I have completed this series of projects, especially because it was never supposed to happen this way. When I started this journey, I was not even sure where it would end up, as I had no idea it would eventually become a four-project series. At the beginning, I was simply focused on learning Artificial Intelligence. Very soon, I realised that to understand AI properly, I first had to understand Machine Learning and Natural Language Processing. My original plan was to build everything using Google Vertex AI, and I thought that once I provided the data, it would automatically build the models and allow me to focus on analysing the results.

But Google Vertex AI never worked on my Google Cloud account. I still don't know what the exact problem was, but I spent two or three days trying different solutions without any success. At one point, I had almost given up on learning Machine Learning altogether. I even considered joining an online course, something I had always tried to avoid because many of them are expensive and often focus more on theory than practical understanding. Python-based Machine Learning was always an option, but I did not want my learning journey to become completely code-oriented. I have never wanted to become a programmer. One of the things I like about AI is that it allows me to focus more on analysis while using it as an assistant for coding rather than making coding the centre of the work.

That was when BigQuery SQL entered the picture. At that time, I was not even sure whether it would help me, so I first started learning the basics of SQL. Surprisingly, I found SQL much easier to understand than Python. Then there was one moment when I finally started understanding the concept of SQL Views. It felt like one of those rare moments where everything suddenly started making sense. I had always built my projects in Excel, where they were dynamic in nature, but I could never control the complete workflow through proper checkpoints. With SQL Views, I realised that every major analytical decision could have its own checkpoint, and those checkpoints could be connected together to build one complete analytical workflow.

At that stage, I had never thought about reusability or reusable systems because that phase did not even exist in my mind. I also had no idea how the Machine Learning Project would finally be structured. But after understanding SQL Views, I completely

changed the way I designed the project. Instead of building isolated queries, I started building connected analytical layers. Looking back now, I think this became one of the biggest turning points of the entire journey.

After that, I started building the Machine Learning Project using BigQuery SQL and SQL Views. The project was mentally exhausting, especially during the model evaluation stage, but every new model gradually added another level of understanding. I moved from Regression to Logistic Classification and finally to the Ensemble models. The transition between the models felt natural because every model extended the understanding developed from the previous one instead of replacing it. In the end, the complete Machine Learning Project was built using SQL Views as analytical checkpoints rather than relying on one large static structure.

When the NLP Project started, another unexpected challenge appeared. BigQuery SQL did not support the NLP functions that I needed. This time I was not as disappointed as before. I was definitely frustrated, but by then I had started believing that if one approach did not work, another solution probably existed. Using Python as an external processing layer created a completely different architecture for the NLP Project while still allowing BigQuery to remain the central analytical platform. Looking back now, I realise that this challenge helped me understand that different types of data require different analytical workflows, and that one solution cannot always be applied to every problem.

After completing both projects, I realised that the SQL Views I had created could do much more than simply organise queries. They could actually make the complete analytical workflow reusable. That understanding became the foundation of the Reusable Systems Project. The objective was no longer only to build Machine Learning and NLP projects, but to make both of them capable of processing new compatible data without rebuilding the complete workflow every time.

When I added the new data to the Machine Learning Project, the reusable workflow was tested for the first time with unseen compatible data. Instead of making me worried, the updated results actually gave me confidence that the project had been built correctly. The complete workflow remained connected, the analytical structure remained stable, and the updated results were generated successfully without changing the overall framework.

The NLP Project also brought its own set of challenges, especially because it worked with unstructured financial news instead of regular structured datasets. Some manual adjustments were still required, particularly in Theme Analysis, and those adjustments were reasonable because they were part of understanding language rather than simply processing numbers. Even with these challenges, the reusable workflow continued to

work successfully, and the overall analytical understanding of the NLP Project remained stable after adding the new data.

In the end, the append analysis gave me much more confidence in the complete framework that I had built. The Machine Learning results remained highly stable, while the changes observed in the NLP Project were also understandable because financial news naturally changes over time. Going through every model and every NLP signal one step at a time helped me understand not only how they work, but also why they behave differently and what each of them contributes towards understanding the data.

Looking back now, I realise that if Google Vertex AI had worked on the very first day, I would probably never have discovered SQL Views, never built this reusable analytical framework, and never understood Machine Learning and Natural Language Processing in the way I do today. What first looked like a failure eventually became the biggest turning point of the entire journey.

More importantly, this journey taught me something beyond Machine Learning and Natural Language Processing. It taught me how to approach a completely new problem, understand the data, build an analytical workflow, interpret the results, and continuously improve the structure whenever I found a better solution. Today, I feel much more confident about approaching a completely new dataset because I now understand how to think through the analytical process instead of simply applying models.

Looking back now, I also realise that this project is not really an ending. It is more like the completion of one stage of my learning. The framework that I have built can continue to evolve whenever new market data becomes available, and many improvements can still be made in the future. But instead of starting from scratch every time, I now have a structured analytical foundation that I understand well. More than anything else, that gives me confidence about working on future analytical problems.

There is a quote that says, *"Sometimes the path that doesn't work becomes the reason you discover a much better one."* Looking back at this complete journey, I feel that this is exactly what happened to me. If one path had worked exactly as I expected, I might never have discovered a much better way of building projects. More than the models, the signals, or the reusable framework itself, this complete journey taught me how to think analytically, how to keep improving whenever I faced a challenge, and how every unexpected problem can become the beginning of something much better.
